Firm Inertia in the Public Cloud *

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ABSTRACT: We document substantial inertia in firms' adoption of public-cloud products, even when switching occurs within the same provider, data center, and product series, where conventional technical and contractual frictions are minimal. Using proprietary customer-level data, we develop and estimate a multiple discrete-continuous demand model of product choice and usage. In counterfactuals holding prices fixed, inertia delays adoption of superior technologies and suppresses cloud usage, reducing consumer surplus by 62% and provider revenue by 58%. Nevertheless, customers derive substantial surplus from cloud use. The findings reveal organizational frictions in firms' adoption of productive inputs despite the cloud's technological flexibility.

Keywords: inertia, firm input demand, cloud computing, multiple discrete-continuous choices

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1 Introduction

Third-party cloud computing services ("the public cloud") have become a central input into modern production. The public cloud allows firms to access computing resources without owning or maintaining physical hardware. By virtualizing infrastructure and charging on a pay-as-you-go basis, cloud providers enable firms to scale usage up or down quickly and to adopt new technologies without large upfront investments. In principle, these features should facilitate rapid reallocation of computing resources toward more efficient technologies as they become available and improve the efficiency with which firms use capital and inputs.

Yet industry evidence suggests that many firms fail to take advantage of these features to use cloud computing resources efficiently.[1] Even as cloud providers regularly introduce products with better performance *and* lower prices, customers appear slow to adjust their product choices. Despite the central role of cloud computing in the modern economy, there is little empirical evidence quantifying the extent of such inertia in firms' cloud demand or assessing its welfare implications.

This paper documents and quantifies inertia in firms' demand for cloud computing products. Using detailed customer-product-level data from a major cloud provider, we show that firms are slow to adopt newly introduced cloud products, even when these products dominate existing ones in terms of price and performance. We find that firms' past product choices strongly affect their future adoption decisions, leading many cloud customers to remain on inferior technologies for extended periods.

We study product adoption in a "best-case" environment for switching: firms choose among products from the same cloud provider, within the same data center, and within the same product series. In this setting, many commonly cited sources of switching frictions—such as data migration costs, interoperability issues, and migration fees—are minimal or absent. As a result, observed inertia is unlikely to be driven by technical barriers or fees imposed by cloud providers. We interpret our results as identifying a lower bound on inertia in digital input markets. If firms exhibit substantial inertia even when switching is technologically straightforward, then inertia is likely to be at least as important in environments where switching costs are higher.

A long literature has documented inertia in *consumer* markets, including health insurance, financial products, packaged goods, and subscriptions (Hortaçsu and Syverson, 2004; Dubé, Hitsch and Rossi, 2010; Handel, 2013; Einav, Klopack and Mahoney, 2025).

[1] Andreessen Horowitz suggests that $100B of market capitalization is lost across the top 50 public software companies due to inefficient cloud usage. See https://a16z.com/the-cost-of-cloud-a-trillion-dollar-paradox/, accessed Apr. 1st 2025.

It is far less clear that organizations exhibit comparable inertia when the stakes are large and adoption is routinely intermediated by specialists. Many conventional switching costs for firms are technical or contractual (Greenstein, 1993; Farrell and Klemperer, 2007), and as discussed, minimal by construction in our setting. The persistent stickiness we find suggests that when input choices are delegated to internal departments, organizational frictions can lead firms to "set it and forget it", generating inertia of a magnitude comparable to that documented for retail consumers, even when many conventional costs of switching are minimal.[2] This also implies that cloud customers would likely remain inert even if regulators banned egress fees and standardized APIs across providers.[3]

Identifying inertia in firms' decision-making is challenging. Sticky product choices may reflect organizational frictions, but they may also arise from persistent unobserved preferences. Distinguishing between these explanations requires variation that exogenously shifts firms' past demand but is uncorrelated with current demand.

We exploit the introduction of new cloud computing products for such variation. The cloud provider periodically introduces new virtual machine (VM) products that offer the same or better performance at lower prices than existing products. Our key empirical strategy compares firms that begin using the cloud just before versus just after a new product launch. Firms that arrive just before a launch necessarily start on older products and may develop inertia, while firms that arrive just after the launch face the same choice set but have no prior usage history. Using a regression discontinuity design (RDD) around product launch dates, we show that firms entering just after a launch are significantly more likely to adopt the new product than otherwise similar firms entering just before. This discontinuity provides causal evidence of inertia, separating it from persistent unobserved preferences.

To quantify the welfare implications of inertia, we develop and estimate a structural demand model tailored to the cloud computing market. A central challenge in this setting is that firms typically use multiple cloud products simultaneously and choose usage levels continuously. Standard discrete-choice models would therefore misrepresent substitution patterns and price sensitivity. Our model allows firms to allocate multiple computing tasks across products endogenously based on prices, product characteristics, and task sizes. Preferences are allowed to be heterogeneous both at the customer and task level, i.e., different

[2]Our interviews with practitioners suggest that once cloud configurations are set, IT teams may lack incentives or capacity to continuously re-optimize usage. Budgeting practices, coordination costs across teams, and limited managerial attention are all organizational frictions that contribute to this inertia. We do not take a stand on what fraction of these frictions are real costs, and we will show that our welfare conclusions are robust to this distinction.

[3]E.g., see https://digital-strategy.ec.europa.eu/en/factpages/data-act-explained, accessed Feb. 9th 2026.

products may be better suited for different customers or different tasks for the same customer.

Inertia is modeled as an upfront cost that firms overcome to adopt a new product, capturing the "set-it-and-forget-it" pattern and organizational frictions. The model is disciplined by the same variation used in the reduced-form analysis, including moments derived from the RDD. This ensures that the estimated inertia is grounded in causal evidence rather than driven by functional-form assumptions. We further show that the multiple-choice continuous-usage model is identified given customer-product-month level data, where researchers observe only the total usage of each customer on each product, but not the number of tasks or the size of each task.

Although the technical barriers to switching are minimal in our setting, firms may still face small residual costs—engineering effort for code changes or brief downtime during migration. Appendix D estimates an extension that allows for such a switching cost in addition to the adoption cost, and finds broadly consistent results: the adoption cost remains the primary friction.

Using the estimated model, we assess how inertia affects firms' product choices and usage. We show that inertia substantially delays adoption of newer products and leads firms to rely more heavily on older, less efficient options. We estimate greater inertia for large firms than for small firms. This is consistent with our interpretation of inertia as organizational frictions, as large firms likely have deeper organizational hierarchies. If the inertia were mainly due to technical barriers, one might expect it to be greater for small firms which typically have less engineering capacity.

We also use the model to summarize the economic implications of these distortions, including their impact on firms' surplus and providers' revenues. These calculations are made under partial-equilibrium, with the supply-side fixed, and they are intended to convey the magnitude of the inefficiencies implied by the inertia.

We find that inertia has large welfare consequences. It reduces firms' consumer surplus by 62 percent relative to a counterfactual without inertia. This loss is overwhelmingly driven by misallocation towards worse computing products rather than the adoption cost itself, so this welfare conclusion is robust regardless of whether inertia is counted as disutility. Surprisingly, inertia also substantially harms the cloud provider: by slowing adoption of newer products that induce usage expansion, inertia reduces provider revenue by 58 percent. This contrasts with many common settings in which inertia benefits incumbents by softening competition (MacKay and Remer, 2024; Einav, Klopack and Mahoney, 2025).

Despite these losses, cloud usage still generates substantial surplus for firms, with av-

erage returns on investment (i.e., consumer surplus divided by cloud spending) exceeding two-to-one. The welfare cost of inertia therefore reflects unrealized gains rather than losses relative to outside options such as private IT infrastructure. Furthermore, small firms, generally faced with tighter budgets and greater demand uncertainty, benefit more from cloud usage in terms of returns on investment.

We then evaluate a counterfactual in which an automatic migration service handles product evaluation and migration on customers' behalf, removing the organizational friction that constitutes inertia. Similar services are already offered in various forms by cloud providers and specialized third-party vendors. We find that the policy generates at least two dollars of total welfare for every dollar spent, even when the service is conservatively priced at the full estimated adoption cost. Because such a service can reuse tooling across many customers, and to the extent the friction is organizational rather than a real resource cost, the actual cost is likely well below this estimate.

Our data comes from a single major cloud provider, but several features of the cloud market suggest the magnitudes we estimate are representative across the major providers. The three largest players in the market, AWS, Azure, and Google Cloud, offer closely substitutable products with matched specifications and similar list prices. The mechanism we identify has no reason to be specific to one provider's institutional setting, so we view our magnitude as a benchmark estimate for the cloud market as a whole.

This paper contributes to the literature on inertia and frictions in choice by documenting substantial inertia in firms' input demand. While most existing evidence on inertia focuses on consumers (Madrian and Shea, 2001; Hortaçsu and Syverson, 2004; Dubé, Hitsch and Rossi, 2010; Shcherbakov, 2016; Chetty et al., 2014; Handel, 2013; Handel and Kolstad, 2015; Brecko, 2023; MacKay and Remer, 2024; Einav, Klopack and Mahoney, 2025), we show that firms exhibit strong persistence in their choice of cloud products, even when newer technologies dominate existing ones. This extends the study of inertia to organizational decision-making and highlights the role of internal frictions in shaping firms' allocation of productive inputs (Cyert and March, 1963; Liebman and Mahoney, 2017). Moreover, in our setting, inertia manifests more as within-firm misallocation across products from the same provider and less as a source of market power for the provider, leading to losses in both firm surplus and provider revenue.

We treat cloud customers as myopic—a standard modeling choice in the inertia literature (Handel, 2013; Einav, Klopack and Mahoney, 2025) and one that aligns with our interpretation of inertia as an organizational "set-it-and-forget-it" friction. An alternative view is that customers are forward-looking and anticipate their own inertia when choosing products (Gowrisankaran and Rysman, 2012; Shcherbakov, 2016; Brecko, 2023). Direct

testing between these views is difficult because it requires identification of the discount factor (Magnac and Thesmar, 2002). One suggestive piece of evidence favors the myopic interpretation in our setting. When the launch of a dominant new product was pre-announced, forward-looking customers who anticipated developing inertia on inferior old products would have an incentive to delay joining the provider until the launch of the dominant new product. However, around one such pre-announced product launch, we detect no slowdown in the growth of cloud usage or the provider's customer base (Figure F3), consistent with our myopic assumption.

Second, we contribute to the emerging literature on the economics of cloud computing (Bloom and Pierri, 2018; Byrne, Corrado and Sichel, 2018; Fang and Greenstein, 2025; Biglaiser, Crémer and Mantovani, 2024). Much of the existing empirical work studies cloud adoption using surveys or census data and focuses on downstream firm outcomes (Jin and Bai, 2022; DeStefano, Kneller and Timmis, 2025; Jin and McElheran, 2025; Caldarola and Fontanelli, 2024; Lu, Phillips and Yang, 2024). We instead study the public cloud market itself, using proprietary data on prices and quantities to analyze firms' product adoption and usage decisions. By focusing on adoption within the same provider, data center, and product series, we examine a setting in which many standard technical switching costs are minimized. Closest to our setting, Brand et al. (2024) document inefficiencies in firms' cloud usage using CPU utilization data and attribute them to learning over time; we complement this perspective by studying product choice and usage decisions, and showing that inertia constitutes a distinct source of inefficiency.

Third, we develop a demand framework that accommodates multiple discrete product choices and continuous usage. In cloud computing and other usage-based markets, firms typically rely on several products simultaneously and adjust usage intensity in response to prices and performance. Standard discrete-choice models, which assume the purchase of a single unit of one product (Berry, 1994; Berry, Levinsohn and Pakes, 1995), abstract from this intensive margin and can substantially misstate substitution patterns and price sensitivity. While a small literature allows for multiple discrete choices (Burda, Harding and Hausman, 2012; Hendel, 1999), these approaches typically model discrete counts rather than continuous usage. Our framework accommodates continuous usage without requiring the econometrician to observe the number or size of underlying tasks, allowing us to quantify inertia and welfare in settings where usage decisions are central. As such, the model is applicable to a broad class of digital and service markets, including cloud computing, large language models, payments, and firm procurement.

Finally, we contribute to the literature measuring the economic benefits of information technology adoption (Brynjolfsson and Hitt, 2000; Bresnahan, Brynjolfsson and Hitt,

2002). Building on classic demand-based approaches to measuring IT returns (Bresnahan, 1986; Hendel, 1999), we structurally estimate demand and usage in the public cloud market to quantify the surplus generated by cloud computing. We show that, despite substantial gains on average, a large fraction of potential benefits remains unrealized due to inertia. This pattern suggests that organizational frictions can significantly dampen the productivity gains from new technologies, even when adoption is privately profitable.

The rest of the paper is organized as follows. Section 2 overviews the public cloud market and describes our data. Section 3 presents descriptive evidence of inertia. Section 4 develops the demand model and discusses identification. Section 5 details the estimation procedure. Section 6 presents the demand estimates. Section 7 conducts counterfactual analyses. Section 8 concludes.

2 Industry Background and Data

2.1 The Public Cloud Market

The public cloud comprises three markets: Infrastructure-as-a-Service (IaaS, e.g., VMs), Platform-as-a-Service (PaaS, e.g., App Services), and Software-as-a-Service (SaaS, e.g., Office 365). This paper focuses on the IaaS market, specifically VMs, which serve as the infrastructure for the rest of the cloud services and digital economy. The IaaS market has four major providers globally, Amazon Web Services, Microsoft Azure, Google Cloud Platform, and Alibaba Cloud, together accounting for almost 80% of the market (Gartner, 2022). These providers operate large data centers and equip them with clusters of computing hardware purchased from upstream component makers such as Intel, AMD, and Nvidia. These clusters, which bundle CPU, RAM, storage, and network, are rented as VMs to customers on an hourly basis using virtualization technology. There are three purchase models of VMs: on-demand, by reservation, and preemptible. We focus on the most popular model: on-demand. As the name suggests, customers can start or stop an on-demand VM at any time with a high performance guarantee and pay based on the amount of computing resources used.[4]

Cloud customers, mostly firms, purchase and configure VMs to substitute or complement traditional private IT infrastructure such as PCs and private data centers. The cloud allows these firms to minimize upfront setup costs and reduce maintenance costs, as they no longer need to purchase and manage physical machines. The cloud also utilizes auto-

[4]In contrast, VMs by reservation typically require one or three years of commitment in exchange for an hourly discount. Preemptible VMs do not have any performance guarantee.

scaling technology, which helps these firms automatically scale the amount of computing resources they use based on their fluctuating demand in real-time. As a result, cloud customers can afford the adoption of the latest technologies in computing hardware and achieve higher utilization of the computing resources they purchase.

2.2 Data

Our data comes from a major cloud provider headquartered in the U.S. and includes prices and characteristics of each VM stock-keeping unit (SKU), along with 32 months of customer-SKU-month level usage history from November 2015 to June 2018. For confidentiality, we retain a random subsample of customers and re-scale the usage and price data each by an unknown constant. Due to this re-scaling, consumer welfare and provider revenue computed in this paper are not interpreted as real-world currency values. Instead, we focus on ratios of welfare measures as well as price elasticities, which retain the same interpretation despite the re-scaling. Furthermore, we hold out the first six months of our data to initialize the usage history of customers who exist since the beginning of our sample, i.e., to define whether a product is new to a customer. This leaves us with 26 months of data for estimation. Finally, when estimating the structural model, we drop transient customers with less than 6 months of usage, leaving us with a total of 3,233 customers in our estimation sample who account for 90% of the total usage.[5]

A SKU is defined by the provider hierarchically as follows: (i) region: location of the data center where customers' workloads would be hosted; (ii) operating system (OS): Linux or Windows; (iii) VM series: product category defined by function, e.g., Azure's D series includes SKUs optimized for CPU-intensive applications; (iv) VM family: group of SKUs within a VM series defined by the underlying hardware, e.g., in Azure's D series, the 3rd generation Dv3 family is based on Intel's Broadwell chips and the 2nd generation Dv2 family is based on the older Haswell chips; (v) VM size: SKUs within a VM family with specified number of virtual CPU cores, gigabytes of RAM and storage, and other resources, e.g., the D2 v3 SKU is in the Dv3 family and comes with 2 virtual CPU cores, 8 gigabytes of RAM, and 50 gigabytes of temporary storage.

We define a region-OS combination as a market, because most customers consistently choose SKUs within the same region and OS. There are 47 such markets in our sample, which are used in all of our analyses throughout.[6] We define a VM family as a product,

[5]These short-lived customers are only excluded from the structural estimation, where they introduce noise because they use the cloud for purposes different from regular customers, such as testing out the provider for initial adoption. We keep these customers in the descriptive evidence in Section 3 for completeness. Table F13 shows that our RDD result is also robust to excluding these customers.

[6]There is little substitution across regions, possibly due to data sovereignty regulations, latency, or simply

grouping VM sizes within a family as one product because they share the same technology and the same price per unit of computing resource (e.g., D4 v3 has twice the CPU cores, RAM, and storage of D2 v3 and costs twice as much per hour). In this paper, we use "product" and "VM" interchangeably. We further focus on the most popular VM series (henceforth X series for confidentiality) and group the VM series other than X as one product named "Other". X series is designed for compute-intensive workloads based on Intel CPUs and accounts for 42% of all usage from this provider. Other VM series are optimized for different workloads, such as memory-intensive, high performance, or GPU-based computing. There are four products in the X series: X1, X2, X3, and X4, from the oldest to the latest. Scripts and integrations built for earlier products in the series typically run unchanged on later products, so there is minimal technical switching cost to move to the later products. X1 and X2 are available throughout our sample, while X3 and X4 are launched during our sample. X2 is launched about one year after X1, X3 about two years after X2, and X4 about three months after X3, with slight variations in launch dates across markets.

Table 1 summarizes the characteristics of the X-series products. For confidentiality, all numeric characteristics are presented as ratios relative to those of X1. For processor features, the maximum clockspeed of X2, X3, and X4 are 1.3, 1.3 and 1.5 times that of X1 respectively, due to newer generations of CPUs. Products after X1 also come with Intel Turbo Boost, which allows their CPUs to run at a clockspeed higher than their base clockspeed when handling heavy workloads. In addition, X4 comes with hyperthreading that allows a single processor to process multiple tasks simultaneously. Hyperthreading, however, does not increase physical resources and slightly reduces single-thread performance for some applications, thus not benefiting all customers. Besides these processor features, the newer products also offer more CPU cores, more memory per core, and faster access to remote resources—both faster remote disk speed and higher bandwidth per core.

X2 and X3 have identical specifications, so there is almost zero technical switching cost between X2 and X3. X3 was launched as a promotional version of X2, two years after X2's launch, with the same features and an average price 24% cheaper. The provider introduced X3 as a separate product—rather than cutting X2's price—to boost demand for the underlying hardware while limiting margin loss on existing X2 customers. X2 remained available after X3's launch, and X3's price had stayed low for 17 months before it was raised to match X2's, after our sample period. During this period, both products were marketed through similar channels, and X3 customers were informed from the beginning that they would not need to switch products when the promotion ended. Still, despite

where the bulk of customers' data are stored.

Table 1: X Series Product Characteristics

	Product	X1	X2	X3	X4
Processor Features	Intel Turbo Boost	No	Yes	Yes	Yes
	Max Clockspeed Compared to X1	1.0	1.3	1.3	1.5
	Hyperthreading	No	No	No	Yes
Available No. Cores Compared to X1	Virtual	1.0	2.0	2.0	8.0
	Physical	1.0	2.0	2.0	4.0
Available Resources per Physical Core Compared to X1	Memory	1.0	1.0	1.0	2.3
	Temp Disk Size	1.0	1.0	1.0	1.0
	Temp Disk RR IOPS	1.0	1.0	1.0	1.0
	Temp Disk RR Speed	1.0	1.0	1.0	1.0
	Temp Disk RW Speed	1.0	1.0	1.0	1.0
	Remote Disk	1.0	1.0	1.0	1.0
	Remote Disk IOPS	1.0	1.0	1.0	1.0
	Remote Disk Speed	1.0	1.5	1.5	1.5
	NICs	1.0	1.0	1.0	1.0
	Bandwidth	1.0	1.5	1.5	2.0
	Average Price	0.577	0.473	0.360	0.469

Notes: Table shows characteristics of products in X-series. For confidentiality, most characteristics are presented as ratios relative to X1, e.g., X2 has twice as many virtual cores as X1. *Max Clockspeed* is the maximum rate at which a computer's CPU executes instructions, measured in hertz (Hz). *Memory* is the maximum memory allocated to a virtual machine, measured in Gibibytes. *Temp Disk Size* is the temporary (local) storage disk size, measured in Gibibytes (GiB). *Temp Disk RR IOPS* is the temporary (local) disk random read input/output operations per second. *Temp Disk RR Speed* is the temporary (local) disk random read speed, measured in megabytes per second. *Temp Disk RW Speed* is the temporary (local) disk random write speed, measured in megabytes per second. *Remote Disk* is the number of remote (uncached) storage disks allowed. *Remote Disk IOPS* is the remote (uncached) disk input/output operations per second. *Remote Disk Speed* the remote (uncached) disk speed, measured in megabytes per second. *NICs* is the number of Network Interface Cards, which allows the VM to communicate with the internet, the cloud provider, and other resources. *Bandwidth* is the sum of all outbound traffic across all network interfaces attached to a virtual machine, measured in megabits per second.
Three variables are presented as is. *Hyperthreading* indicates whether the products are equipped with the hyperthreading technology, which allows a single processor to run multiple threads simultaneously. *Intel Turbo Boost* indicates whether the products are equipped with the Intel Turbo Boost technology, which allows the CPU run at its base clock speed when handling light workloads, then jump to a higher clock speed for heavy workloads. *Average Price* is the average across product-market-months and expressed in re-scaled dollars per compute unit.

the identical underlying hardware and the provider's efforts, X3's promotional status may create a perception of lower quality, which we will discuss in the next section.

Prices are measured in re-scaled dollars per compute unit, where compute units measure the amount of computing resources consumed. Cloud providers standardize computing resources across products using compute units, taking into account both the number of

core-hours used as well as the performance of each core across different products.[7] Table 1 shows that the average price declines in newer generations of products (except the promotional X3). Taken together, we see that new products in the cloud market typically come with better technology, more resources, and lower prices, while old products continue to be available in the market.

Moreover, prices vary mainly across markets (i.e., region-OS) and products.[8] The most expensive region is 56% more expensive than the cheapest region on average, and Windows VMs have a price premium over Linux VMs. Within an average region-OS-month, the most expensive X-series product is priced 35% higher than the cheapest X-series product. We leverage the variation across regions and products to identify cloud customers' price elasticities. To this end, we supplement our data with three cost shifters obtained from the cloud provider: (i) electricity prices across regions; (ii) power consumption ratings across product-regions; and (iii) hardware costs across product-regions. Table F7 presents additional summary statistics on usage and prices, and Appendix B.1 provides details on the cost shifters.

Finally, we supplement the provider's data with industry estimates of the total market size to infer customers' computing needs outside the provider, which are fulfilled by private IT infrastructure and competing public providers. Cisco (2018) and Gartner (2017) estimate that the public cloud accounts for 58% of total cloud usage, and cloud usage accounts for 83% of total computing workloads worldwide. We therefore construct the outside option share as one minus the product of these shares and the provider's share of the public cloud market.[9] Most customers used a single public cloud provider during our sample period, so the outside option for them is primarily private IT infrastructure rather than competing public clouds.[10]

[7]The number of core-hours used by a VM is calculated by multiplying the VM's number of CPU cores with the number of hours used. As the performance of each CPU core may differ for different VMs, cloud providers use compute units to normalize core-hours so that usage across different VMs is comparable. For example, see https://docs.microsoft.com/en-us/azure/virtual-machines/acu for Azure compute unit, accessed Dec. 26th 2021.

[8]Over time, there was only one major price drop where prices for X2 are marked down in about one third of the markets at the same time.

[9]This estimate is necessarily approximate and our main results do not rely on it. It only enters the structural estimation to pin down the usage of the outside option, and most of our findings are robust to alternative values.

[10]Annual industry surveys by RightScale (now Flexera) indicate that in 2018 only about one quarter of public cloud customers use more than one public cloud provider. See https://info.flexera.com/CM-REPORT-State-of-the-Cloud, accessed Feb 6th 2026.

3 Descriptive Evidence

We start by presenting three pieces of evidence showing that cloud customers develop inertia towards products they have previously used. Then, we present usage patterns that motivate our demand model.

3.1 Inertia

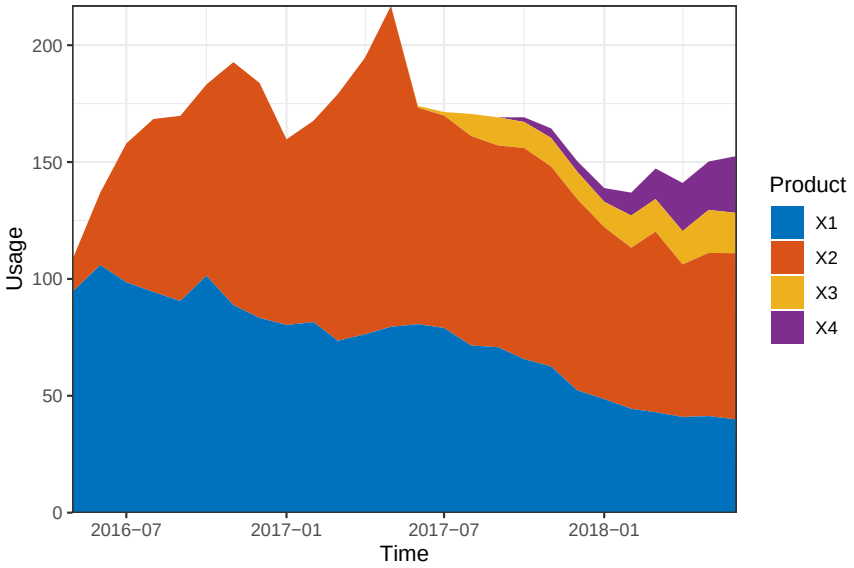
In our "best-case" switching environment, where the technical and contractual sources of switching costs are minimal by construction, we interpret the inertia we document as primarily reflecting organizational frictions. Our interviews with industry practitioners suggest that cloud adoption decisions are typically delegated to IT departments, which face few incentives to continuously revisit established configurations; reducing cloud spending may even cut into the IT department's own budget. Configurations therefore tend to be revisited only when triggered by an external shock, such as a finance-driven cost cut, pressure from engineering to improve performance, or a new task that the existing configuration cannot serve.

Therefore, we measure inertia in terms of the cost of adopting a product that is new to the customer. We show that new product adoption is slow and affected by customers' past product choices, building our evidence from descriptive patterns to causal findings under plausible identifying assumptions. These results suggest that inertia leads customers to significantly overpay for computing. Meanwhile, they also motivate our demand model, which we use to interpret the magnitude of inertia, understand its welfare impact, and evaluate counterfactual policies.

Slow adoption of new products. Because new cloud products typically dominate old ones with better technology, more resources, and lower prices, slow adoptions of new products by cloud customers could indicate the presence of inertia. To illustrate, Figure 1 plots usage of the X series products over time for a panel of customers who exist since the beginning of our sample. These customers are slow to adopt the new products, X3 and X4. At the end of our sample, X3 and X4 only account for 27% of their total X series usage, 13 months and 9 months after their initial releases, respectively. Moreover, these customers are also slow to migrate off X1, which still accounts for 26% of the total usage at the end of our sample despite its high cost and low performance. Figure 1, however, is not sufficient to show that inertia is present, as serial correlation in unobserved preferences may generate the same "sticky" usage pattern. Therefore, we now compare

different cohorts of customers, who adopted the cloud provider at different times during our sample, to illustrate how their past choices impact their adoptions of new products.

Figure 1: Usage of initial customers over time within X series



Notes: Figure plots usage over time by product for customers who adopted the cloud provider before the start of our sample.

Impact of past product choices on new product adoptions. When firms first become customers of the provider, they have no usage history on any product, and thus no inertia. Over time, they may develop inertia and face higher costs of adopting new products. Thus, comparing old and new customers reveals inertia by showing how past choices affect new product adoptions.

Specifically, we conduct an analysis of different customer cohorts' product choices similar to Handel (2013). We divide the sample into three periods: (a) before the launch of X3, (b) between the launches of X3 and X4, and (c) after the launch of X4. Correspondingly, we have three cohorts of customers based on the periods in which they first become customers of the provider, denoted as cohorts A, B, and C. For example, customers in cohort B first become customers of the provider after the launch of X3 but before that of X4. If inertia is present and these customer cohorts are otherwise similar, then cohorts in periods when they are new should be more likely to adopt the dominant new products than the older cohorts who have developed inertia.

Table 2 reports usage shares across the X-series products for the three customer cohorts in each of the three time periods. Cohort A—customers who joined the provider before X3 launched—initially split their usage between X1 (45%) and X2 (55%) in period (a).

When X3 became available in period (b), only 2% of Cohort A's usage moved onto the new product, compared to 16% for the new Cohort B that first joined after X3's launch. This contrast reflects the impact of past choices on new product adoption. Even Cohort B's 16% adoption of X3 is low given that X3 is the same product as X2 at a lower price; we attribute this to a perceived quality discount on the promotional version (similar to Bronnenberg et al. 2015) and account for it in our structural model. The pattern strengthens after X4's launch in period (c): 30% of usage by the new Cohort C goes onto X4, compared to 9% for Cohort A and 5% for Cohort B. Within each cohort, usage shares are also sticky over time. Usage on older products declines slowly while usage on new products grows slowly, indicating the presence of inertia. Table F8 shows similar patterns in a balanced panel, which rules out attrition as the driver.

Table 2: Customer cohorts' usage shares within X series

	(A) Before X3	(B) New Customer Between X3 to X4	(C) New Customer After X4
(a) Before X3 Usage Share			
X1	45%	-	-
X2	55%	-	-
(b) X3 to X4 Usage Share			
X1	28%	14%	-
X2	71%	71%	-
X3	2%	16%	-
(c) After X4 Usage Share			
X1	22%	11%	10%
X2	63%	65%	54%
X3	6%	19%	7%
X4	9%	5%	30%

Notes: Table presents usage share across products within X series for three customer cohorts and three time periods. The three cohorts (presented in columns) are (A) customers who adopted the cloud provider before the launch of X3, (B) customers who adopted the cloud provider between the launch of X3 and X4, and (C) customers who adopted the cloud provider after the launch of X4. The three time periods are (a) before the launch of X3, (b) between the launch of X3 and X4, and (c) after the launch of X4. Panel (a)–(c) present the usage shares in each of the three time periods.

These product adoption decisions represent significant financial stakes for the cloud customers. Take X1 and X4 as an example. The launch of X4 offers a strictly dominating option compared to X1, with better technology, more resources, and lower prices. However, we find that 76% of X1 customers do not adopt X4, thus over-paying on the cloud by over 20%, before even adjusting for the performance differences.

A regression discontinuity design. To identify the causal effect of past product choices on new product adoption, we need to ensure that the cohorts we compare are otherwise similar. Therefore, to sharpen our identification further, we implement a regression discontinuity design (RDD) around each product launch, leveraging our customer-month data.

The running variable is the month in which a customer first joined the provider; the cut-off is the launch month of the new product. Customers who joined just before the launch (*Existing*) and just after (*New*) differ in prior product exposure. *Existing* customers have been exposed to other products and thus may have developed inertia, while *New* customers have no usage history and therefore no inertia. The identifying assumption is that the two groups differ in their past choices but are otherwise similar. Under this assumption, a positive discontinuity in adoption of the new product at the cut-off would serve as evidence of inertia.

Formally, we estimate the following linear probability model on customers who joined the provider within 10 months on either side of the X4 launch (our baseline window):

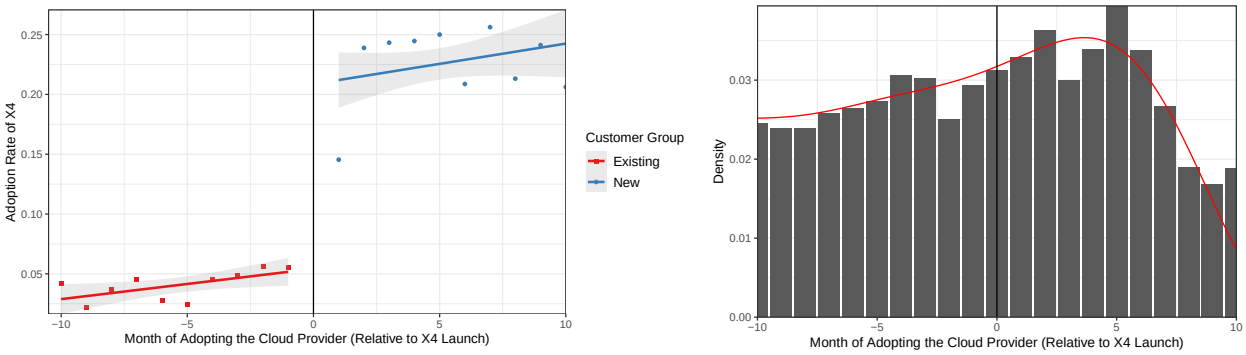
$$\mathit{AdoptNew}_{ij} = \delta_0 \mathit{NewCustomer}_{ij} + \delta_1 T_{ij} + \delta_2 \mathit{NewCustomer}_{ij} \times T_{ij} + \nu_{ij},$$

where $\mathit{AdoptNew}_{ij}$ equals one if customer i adopts product j in the launch month or any of the subsequent 10 months (our baseline adoption horizon), $\mathit{NewCustomer}_{ij}$ indicates whether i joined after j's launch, and $T_{ij} = \mathit{CustomerCohortTime}_i - \mathit{ProductLaunchTime}_j$ is the running variable re-centered at the cut-off. Because X4 is launched mid-month, we use a donut-RDD that excludes the launch month (Barreca et al., 2011; Dong and Kolesár, 2023). The coefficient of interest is δ_0—the discontinuity in adoption at the cut-off—which captures the impact of inertia.

Figure 2: X4 adoption rate after its launch

(a) Regression discontinuity (linear control)

(b) Density of the running variable



Notes: Figure shows a donut-RDD comparing X4 adoption between existing and new customers. The running variable is the month a customer first adopts the provider relative to X4's launch; the outcome is an indicator for adopting X4. We restrict to customers adopting the provider within 10 months before/after the launch, excluding the launch month. Panel (a) plots local regressions on either side with 95% confidence bands. Panel (b) shows the running-variable density (kernel fit in red; histogram rescaled as the probability densities in gray).

Figure 2 shows the result. Panel (a) plots local linear fits on each side of the cut-off. The estimated gap is $\hat{\delta}_0 = 0.15$ (Table F9), which is statistically significant at 1% with

standard errors clustered at the market-running variable level (Lee and Card, 2008). This implies that customers who joined just after X4's launch are 15 percentage points more likely to adopt it than customers who joined just before. Panel (b) shows that the density of the running variable is smooth at the cut-off, suggesting that there is no clear evidence of selection (McCrary, 2008).

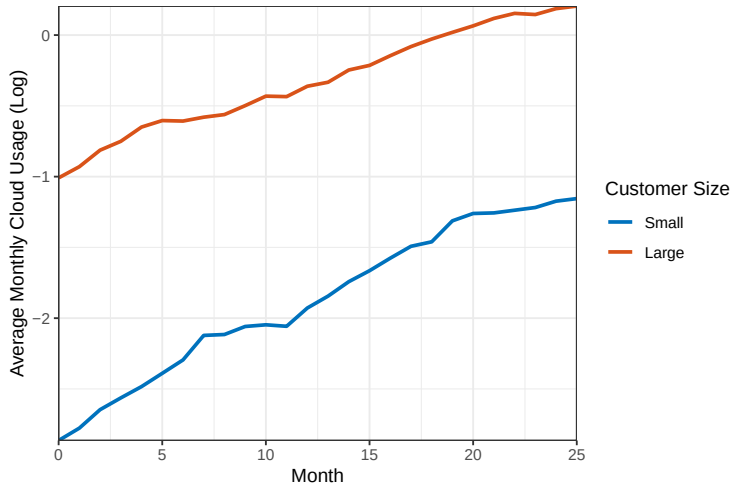
The result replicates at the X3 launch, where we again find a statistically significant positive discontinuity at the cut-off (Table F10, Figure F5). This inertia is unlikely to reflect technical barriers, because X2 and X3 have identical specifications (Table 1).

The RDD results are also robust to alternative specifications of the running-variable control, the estimation window, and the adoption horizon. Under a quadratic control of the running variable, the gap at the cut-off remains positive and statistically significant at both launches (Tables F9 and F10), and the estimate stays positive and broadly stable across all windows and horizons, and statistically significant in all but one X4 adoption horizon (Tables F11 and F12).

These patterns establish that past usage shapes future product choices and that this inertia leads customers to overpay for computing. To fully assess the welfare impact of this inertia, we must also account for performance differences across products. This motivates the need for a unified framework that jointly models price, product quality, and inertia.

3.2 Demand Patterns

Figure 3: Large customers use more cloud but small customers grow faster



Notes: Figure shows how logged average monthly cloud usage changes over time for small and large customers. The horizontal axis presents the number of months since the start of the sample in May 2016. The vertical axis presents the logged average monthly cloud usage in anonymized compute units.

We now document two additional demand patterns that inform our modeling choices. First, usage growth differs by customer size. We classify customers as large if their usage in the first six months is above the median and small otherwise. Figure 3 shows the growth in the cloud usage in log scale for large and small customers. Large customers use five times as much cloud computing as small customers on average, but small customers grow faster, by more than 4x over the sample period versus about 2x for large customers. By the end of the sample, an average small customer uses more cloud than roughly the bottom quarter of large customers. Motivated by these differences, we allow demand parameters in our model to vary by customer size.

Second, cloud usage is continuous and customers often use multiple products within a month. Table 3 reports the joint distribution of the number of products used and total usage per customer-month. Multiple products are used in 17% of customer-months, accounting for 46% of total cloud usage. Customers with higher usage typically use more distinct products, consistent with them facing more heterogeneous computing tasks. These features motivate a demand model that allows multiple product choices within a period, continuous usage, and heterogeneity at both the customer and computing-task level.

Table 3: Frequency distribution of number of products chosen and usage

Usage \ No. Products	1	2	3	4	5	Total
0-0.1	10316	479	73	3	1	10872
0.1-1	38469	4469	523	17	1	43479
1-10	5002	3613	1130	159	18	9922
10-100	63	97	93	17	3	273
Total	53850	8658	1819	196	23	64546

Notes: Table shows the number of customer-market-months tabulated by the number of distinct products chosen (in columns) and the total usage within the same market-month (in rows), excluding customer-market-months with zero products chosen. Usage is measured in re-scaled compute units. Products include X1 to X4, and the "other" product.

4 Model and Identification

We first present a hierarchical demand model that allows for multiple product choices and continuous usage on each product. Then, we discuss the identification of this model. We model the cloud customers as myopic. With a slight abuse of notation, we refer to market-months simply as markets and denote them by m. Customers are indexed by i, and model parameters with subscript i vary by customer size to capture their usage differences and welfare implications.

Each month, cloud customer i faces a random number of computing tasks n_{im}, drawn i.i.d. from a Poisson distribution: $n_{im} \sim Poisson(\lambda_{im})$. The expected number of tasks in market m is denoted by λ_{im} and has a time trend to capture the growing computation needs: $\lambda_{im} = \lambda_{i0} + \lambda_{i1} t_m$. Each task for customer i in market m is endowed with task size q_{im} drawn from an exponential distribution: $q_{im} \sim Exp(\gamma_i)$, where γ_i is the scale parameter and is equal to the expected task size. The task size represents the amount of computing resources needed to complete the task, which is in compute units and thus the same regardless of the product it runs on. Different from Burda, Harding and Hausman (2012), we allow usage to be continuous, so our model applies to any usage-based services.

In the rest of this section, we first describe customer utility and product choice at the task level, and then discuss the aggregation to the product level.

For each task, the customer chooses between J available VMs and the outside option. The customer's utility from using VM j in market m for task k is:

$$u_{ijmk} = \alpha_i p_{jm} q_{im} + X_j \beta_i + \delta_i New_{ijm} + \xi_{ijm} + \epsilon_{ijmk}, \tag{1}$$

where p_{jm} is the per compute unit price for VM j in market m and α_i is the price coefficient; $\alpha_i p_{jm} q_{im}$ measures the customer's dis-utility from the cost of the task. X_j is a vector of VM j's observable characteristics and $X_j \beta_i$ measures the customer's utility from these characteristics. In our setting, X_j is comprised of dummies for each product in the X series and the "other" product.[11] Because X3 is the promotional version of X2, we specify the utility for X3 as the sum of the X2 dummy plus an indicator $Promo_j$, which equals 1 only for X3.

To model inertia, we define New_{ijm} as a new product indicator that equals 1 if customer i has never used product j in market m, i.e., the product is new to the customer. This specification captures the inertia that prevents customers from adopting new products, shown in the descriptive evidence. We call δ_i the adoption cost. In our setting, we interpret δ_i primarily as organizational frictions firms face in adopting a new VM, rather than technical barriers. We do not attempt to separate the component of δ_i that reflects the real cost of internal processes from the component that reflects behavioral inertia; as we show in Section 7.1, our welfare conclusions do not depend on this split.

Demand shocks for customer i using VM j in market m that are unobservable to the econometrician is represented by ξ_{ijm}. These shocks may create endogeneity in prices. We follow Petrin and Train (2010) and address this with a control-function approach,

[11] Even though we observe the detailed product characteristics presented in Table 1, there are not enough products to separately identify the preference for each of these characteristics, so we include the fixed effects instead.

instrumenting for prices using the cost shifters introduced in Section 2; we describe the procedure in Section 5. This procedure will let the coefficients on the product dummies X_j have random coefficients with means varying by customer sizes, while other preference coefficients vary by customer sizes only.

Finally, ϵ_{ijmk} is an idiosyncratic task-level preference shock, distributed i.i.d. type-I extreme value, for all $j = 0, 1, ..., J$. If the outside option is chosen, the customer receives utility $u_{i0mk} = \epsilon_{i0mk}$. We normalize the mean utility of the outside option to be zero, so u_{ijmk} measures the incremental utility from this provider's VM j relative to the outside option.

Putting it together, the task-level choice probability is as follows. Let $d_{ijmk}(q_{im})$ be a dummy variable, which equals one if VM j is chosen for task k, i.e., $u_{ijmk} > u_{ilmk}$ for all $l \neq j$. Then, the probability of choosing VM j for task k of size q_{im} is given by

$$\mathbb{P}(d_{ijmk}(q_{im}) = 1) \equiv P_{ijm}(q_{im}) = \begin{cases} \frac{\exp(\alpha_i p_{jm} q_{im} + X_j \beta_i + \delta_i New_{ijm} + \xi_{ijm})}{1 + \sum_{l=1}^{J} \exp(\alpha_i p_{lm} q_{im} + X_l \beta_i + \delta_i New_{ilm} + \xi_{ilm})} & \text{if } j \neq 0 \\ \frac{1}{1 + \sum_{l=1}^{J} \exp(\alpha_i p_{lm} q_{im} + X_l \beta_i + \delta_i New_{ilm} + \xi_{ilm})} & \text{otherwise.} \end{cases} \tag{2}$$

4.1 Identification

To build intuition, we first discuss the identification of a simplified version of our demand model that omits inertia, customer heterogeneity, and time trend in the number of tasks. Crucially, we will show that the model parameters are identified with customer-level rather than task level data. In the simplified model, there are at most $J + 3$ parameters: up to J parameters for product characteristics (β's), a single price coefficient (α), and two parameters governing the task generation process (λ and γ). We conclude this section by reintroducing the omitted features.

Denote y_{ijm} as the total usage for customer i on product j in market m across all its tasks, i.e., $y_{ijm} = \sum_{k=1}^{n_{im}} q_{im} d_{ijmk}(q_{im})$. To identify the model, we require that for any customer i, its total usage on every product $(y_{i0m}, y_{i1m}, \cdots y_{iJm})$ is observed. This requirement is natural in most markets with continuous usage and more flexible than Burda, Harding and Hausman (2012), as we do not require the number of tasks or the size of each task to be observable. Unlike Burda, Harding and Hausman (2012), the likelihood function for our hierarchical model cannot be easily constructed. This is because our observable y_{ijm} is a continuous variable and, without observing task sizes, its density depends on an infinite sum of convoluted integration (see Appendix A.1 for details). Therefore, we instead use generalized method of moments (GMM) to estimate the model and consider the following two sets of moments for identification.

Proposition 1 (Zero Usage Probability and Expected Usage Moments)**.** *The probability of customer i having zero usage on product j in market m is given by*

$$\mathbb{P}(y_{ijm}=0)=\sum_{n_{im}=0}^{\infty}\underbrace{\frac{\exp(-\lambda)\lambda^{n_{im}}}{n_{im}!}}_{Pr(\text{Number of Tasks})}\cdot\int_{q_{im}}\underbrace{(1-P_{ijm}(q_{im})\ )^{n_{im}}}_{Pr(\text{Product Choice})}\underbrace{\frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im})}_{Pr(\text{Task Size})}dq_{im}; \tag{3}$$

and the expected usage is given by

$$\mathbb{E}(y_{ijm})=\underbrace{\lambda}_{\mathbb{E}(\text{Number of Tasks})}\underbrace{\int_{q_{im}}q_{im}P_{ijm}(q_{im})\frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im})\,dq_{im}}_{\mathbb{E}(\text{Task Size})}. \tag{4}$$

Proof. See Appendix A.2. □

In Proposition 1, the first moment $\mathbb{P}(y_{ijm}=0)$, the probability of zero usage on each product, captures the sparsity of customers' product choices and helps identify the number of tasks. To see how it is calculated, for any given number of tasks n_{im} and task size q_{im}, $\mathbb{P}(y_{ijm}=0)$ has three components, as shown in equation (3): *Pr(Number of Tasks)* is the probability of receiving n_{im} tasks; *Pr(Product Choice)* is the probability of not choosing product j in any of these n_{im} tasks of size q_{im}; *Pr(Task Size)* is the probability of receiving a task of size q_{im}. Multiplying the three probabilities gives the probability of zero usage on a product for a given number of tasks and task size. We then integrate over all task numbers and sizes to obtain the zero usage moment.

To better understand why the sparsity of customers' product choices help identify the number of tasks, consider an extreme case of zero usage on *all* products. The probability that customer i in market m has zero usage on all products is given by

$$\mathbb{P}(\sum_{j=0}^{J}y_{ijm}=0)\equiv\mathbb{P}(n_{im}=0)=\exp(-\lambda).$$

We could simply invert the sample analogue of this probability to directly back out the average number of tasks λ. In practice, we instead use the probability of having zero usage on each product $\mathbb{P}(y_{ijm}=0)$ for $j\neq 0$ as moments. This is because, as introduced in Section 2, we only observe an aggregate share on the outside option usage from the industry reports, so we refrain from directly constructing the sample analogue of $\mathbb{P}(\sum_{j=0}^{J}y_{ijm}=0)$.

The second moment in Proposition 1 is the expected usage on each product $\mathbb{E}(y_{ijm})$ and helps identify task size. As shown in equation (4), product j's expected usage is calculated as the product of the expected number of tasks and the expected task size if product j is

chosen. To see how each product's expected usage helps identify task size, we again take the sum of this moment across all products for customer i in market m:

$$\sum_{j=0}^{J} \mathbb{E}(y_{ijm}) \equiv \mathbb{E}(y_{im}) = \lambda\gamma,$$

which depends only on the task generation process. So if the average number of tasks λ is identified (by the first moment), this sum pins down the average task size γ.

Besides the task generation process, the zero usage probability and the expected usage on each product also help identify preference parameters α's and β's in the utility function, as in standard logit demand models. Together, these two sets of moments jointly identify all parameters in the simplified model.[12]

Finally, going back to our full model, in order to identify inertia, customer heterogeneity, and time trend in the number of tasks, we construct the zero usage probability and expected usage moments conditional on customer and market characteristics. We detail these conditional moments in Section 5.1.

5 Estimation

Following the identification argument, we estimate the model with GMM. We begin by describing all the moments used in estimation. Then, we discuss how we account for potential price endogeneity. Finally, we introduce our estimation procedure.

5.1 Moments

Estimation uses the two moments from Proposition 1—the zero-usage probability $\mathbb{P}(y_{ijm} = 0)$ and expected usage $\mathbb{E}(y_{ijm})$—conditioned on customer and market characteristics to identify inertia, customer heterogeneity, and time trends in computing demand. To sharpen the identification of inertia, we additionally condition the zero-usage probability on whether a customer joined the cloud provider before or after a new product's launch (*NewCustomer*$_{ij}$), exploiting the donut-RDD variation from Section 3. Table 4 lists the full set of moment conditions. The parameters are jointly identified. We discuss the primary source of variation for each below.

[12] In a market with J products and an outside option, Proposition 1 gives $J + 1$ moment conditions for the zero usage probability moments and $J + 1$ moment conditions for the expected usage moments. These $2 \times (J + 1)$ moment conditions are sufficient to parametrically identify the at most $J + 3$ parameters of the simplified model for any $J \geq 1$.

Table 4: Moment conditions

Moments	Characteristics conditioned on	# Moment conditions
Zero Usage Probability	- region, choice set	178
	- *New*$_{ijm}$ (for $j \neq 0$), customer size	20
	- RDD: *NewCustomer*$_{ij}$, customer size	8
Expected Usage	- region, choice set	222
	- *New*$_{ijm}$ (for $j \neq 0$), customer size	20
	- month (t_m), customer size	52

We first condition both moments on all combinations of region and choice set. Conditioning on region helps identify the preference parameters, especially the price coefficient, as prices vary widely across regions. The substitution patterns across different regions also identify the distribution of the unobservable demand shocks ξ_{ijm}. Conditioning on choice set captures variation in the set of products available across markets. For example, the same region's choice set is different after a product launch, so we compute moments before and after the launch separately. Given the different product availability across markets, there are eight unique choice sets in total. We drop moments with few observations or outliers, leaving us with $178 + 222 = 400$ moments.[13]

Second, we condition both moments on whether a cloud product is new to a customer in a market, as well as the size of the customer (i.e., large and small). These moments reflect differences in choice probabilities and usage with and without inertia, thus identifying the adoption cost parameters. Conditioning on customer size helps identify heterogeneity between small and large customers. Combining 5 products, 2 values of New_{ijm}, and 2 customer sizes yields 20 moment conditions per moment type—20 from the zero-usage probability and 20 from expected usage.

To separate the adoption cost from unobserved preference heterogeneity, we exploit the donut-RDD variation in Section 3 by conditioning the zero-usage probability on *NewCustomer*$_{ij}$ (whether the customer joined before or after the new product's launch). We focus on the two newly launched products (X3 and X4) and customer cohorts entering within 3 months of each launch, excluding the launch month itself. Adding customer size, this yields $2 \times 2 \times 2 = 8$ RDD moments.

Finally, we condition total expected usage (summed across products) on month and customer size, identifying the time trend in the number of tasks separately for small and large customers. This adds $26 \times 2 = 52$ moments.

[13]We drop moments with fewer than 200 observations. We also drop one expected usage moment for one region-product where all but 3 out of 311 observations are zeroes.

5.2 Price Endogeneity

A common identification challenge in demand estimation is that prices may be correlated with unobserved demand shocks. In our setting, that is, $p_{jm} \not\perp \xi_{ijm}$. To account for the potential price endogeneity, we use the control function approach with the cost shifters introduced in Section 2 as instruments.

Let z_{jm} denote cost shifters at the product-market level, which affect prices of the corresponding product-markets and are independent of demand shocks ϵ_{ijmk} and ξ_{ijm}. Following Petrin and Train (2010), we assume price is additive in its observed and unobserved covariates, i.e.,

$$p_{jm} = W(z_{jm}, \zeta) + \theta_{jm}, \tag{5}$$

where θ_{jm} is the unobserved covariate that is independent of z_{jm} but correlated with ξ_{ijm}. We decompose $\xi_{ijm} = \kappa_i \theta_{jm} + \tilde{\xi}_{ijm}$, where $\tilde{\xi}_{ijm}$ is a demand shock independent of prices. Then, we rewrite $\theta_{jm} = p_{jm} - W(z_{jm}, \zeta)$ following equation (5). Finally, substituting ξ_{ijm} in equation (1), we derive the utility function with control function as

$$u_{ijmk} = \alpha_i p_{jm} q_{im} + \beta_i X_j + \delta_i \mathit{New}_{ijm} + \kappa_i \left(p_{jm} - W(z_{jm}, \zeta)\right) + \tilde{\xi}_{ijm} + \epsilon_{ijmk}, \tag{6}$$

where $p_{jm} - W(z_{jm}, \zeta)$ is the control function.

5.3 Estimation Procedure

We estimate the model in two stages. In the first stage, we estimate the control function by regressing prices p_{jm} on the instruments z_{jm} and taking the residuals $\hat{\theta}_{jm} = p_{jm} - W(z_{jm}, \hat{\zeta})$. The instruments include electricity prices (varying across regions), power consumption ratings (varying across products and regions), and the provider's hardware procurement costs (varying across products and regions).[14] The construction of the instruments and the first-stage results are detailed in Appendix B.1.

In the second stage, we treat the estimated control function $\hat{\theta}_{jm}$ as a product characteristic as in equation (6). We assume that the price-independent component of the unobserved demand shock, $\tilde{\xi}_{ijm}$, is normally distributed with mean zero and standard deviation σ_j, and allow the standard deviation to vary across products. This is effectively adding random coefficients to the product dummies X_j with means that vary across customer

[14] The advertised performance of a VM in Table 1 can be guaranteed by different combinations of physical hardware using the virtualization technology, so VMs may be supported by different hardware in different regions.

sizes. We also exclude every customer's first-month data from estimation given our focus on cloud customers' product choices and usage rather than their decisions on adopting the provider.

The rest of the second stage follows the standard GMM procedure using moment conditions listed in Table 4. We first compute these moments in the data. Then, for any set of parameter values, we use numerical integration to compute the model moments. Finally, we run a quasi-Newton algorithm to minimize the weighted sum of the squared differences between data moments and model moments. Details of the numerical integration and the optimization algorithm are presented in Appendix B.2. The weighting matrix is a diagonal matrix containing the inverse of the moments' sample variance. We compute standard errors using the GMM asymptotic variance. In Appendix C, we show convergence of this estimator in finite samples with Monte Carlo simulations.

6 Results

Table 5 presents our estimated demand parameters. Panel (a) and (b) present parameters governing product choice, including mean coefficients and standard deviation of the random coefficients. Panel (c) presents parameters governing the distribution of the number of tasks. Panel (d) presents the parameter governing task size distribution. The two columns separately show estimates for small and large customers.

The estimates of the product dummies are in line with the evolution of the cloud products' technical specifications. Quality increases significantly from X1 to X2/X3 due to major improvements of the underlying CPU technology. Meanwhile, the CPU improvement from X2/X3 to X4 is on a minor Intel release, resulting in similar performance between these products.[15] X4 is also launched with hyperthreading, which reduces its cost but slightly hurts its performance. Moreover, consistent with the earlier descriptive evidence, the perceived quality of a promotional product is significantly lower. Note that our estimated product dummies are all negative, because during our sample period, cloud usage still only accounts for a small fraction of total computing needs (Cisco, 2018; Gartner, 2017). Finally, the estimated standard deviations of the random coefficients are statistically significant and sizeable for X1–X4, suggesting significant heterogeneity in preferences for the X-series products across customers and markets.

The price coefficients are negative and statistically significant, i.e., cloud customers

[15] Intel has a tick-tock model of CPU innovation, alternating between major CPU improvements ("tock") and minor cost-reducing die shrinks ("tick") in its product launches. See https://www.intel.com/content/www/us/en/silicon-innovations/intel-tick-tock-model-general.html, accessed Feb. 28th 2021.

Table 5: Demand estimates

Parameter	Estimate	SE	Estimate	SE
Panel (a) Product choice: means	Small		Large	
X1 (β_i^1)	-4.328	(0.115)	-3.460	(0.089)
X2/X3 (β_i^2)	-3.888	(0.106)	-3.174	(0.086)
X4 (β_i^4)	-4.011	(0.133)	-3.105	(0.096)
Other (β_i^O)	-2.608	(2.318)	-2.345	(2.271)
Promo (β_i^P)	-0.851	(0.093)	-0.297	(0.053)
Price $\times$ task size (α_i)	-7.894	(1.539)	-4.215	(0.646)
Adoption cost (δ_i)	-5.486	(0.165)	-6.093	(0.325)
Control function (κ_i)	3.368	(0.491)	0.912	(0.244)
Panel (b) Product choice: standard deviations				
X1 (σ^1)			1.148	(0.106)
X2 (σ^2)			1.016	(0.071)
X3 (σ^3)			1.300	(0.146)
X4 (σ^4)			1.337	(0.101)
Other (σ^O)			0.002	(13.342)
Panel (c) Number of tasks	Small		Large	
Intercept (λ_{i0})	43.208	(2.866)	80.780	(4.807)
Time trend (λ_{i1})	1.849	(0.184)	0.460	(0.131)
Panel (d) Task size	Small		Large	
Exponential scale (γ_i)	0.105	(0.007)	0.371	(0.018)

Notes: Table presents estimates of demand parameters. Standard errors are in parentheses and computed using the estimated GMM asymptotic variance. We allow all parameters to depend on whether a customer is small or large, except for the standard deviations for the random coefficients in Panel (b). In Panel (a), (c), and (d), the first (last) two columns present the parameter estimates and standard errors for small (large) customers.

Table 6: Own-price elasticities

Product	Avg.	Small customers	Large customers
X1	-0.824	-0.672	-0.987
X2	-0.732	-0.586	-0.886
X3	-0.629	-0.488	-0.765
X4	-0.744	-0.589	-0.886
Other	-0.942	-0.789	-1.101

Notes: Table presents estimated own-price elasticities for different products and customer sizes.

dislike higher prices. Conditional on the same amount of computing needs, we find small customers to be more price sensitive than large customers. However, taking into account differences in computing needs yields higher price elasticities for large customers. The own-price elasticity for VM j is computed as the change in expected usage on VM j in response to a unit change in its price, i.e., $\frac{\partial \mathbb{E}(y_{ijm})}{\partial p_{jm}} \frac{p_{jm}}{\mathbb{E}(y_{ijm})}$ (formula in Appendix A.3 and results in Table 6). Cloud customers are generally inelastic to prices. The estimated own-price elasticities are lower than those of most consumer products, though slightly higher than short-run elasticity for electricity (Deryugina, MacKay and Reif, 2020). Our estimated low price elasticities are consistent with other studies in emerging digital markets, where providers are often concerned with long-run growth objectives and do not increase prices despite inelastic demand (Castillo, 2025).

We find that customers face significant inertia when adopting a new product. The estimated adoption cost is equivalent to 17 times the major quality improvement from X1 to X2/X3.[16] These large adoption costs prevent cloud customers from adopting new products, which are typically more powerful and cheaper, thus limiting the potential welfare benefits from cloud. We discuss this point in depth in the counterfactual section. Finally, small customers face lower adoption costs than large customers, consistent with the organizational-friction interpretation: such frictions tend to grow with firm size and organizational complexity.

Turning to the task generation process, we find that small customers have fewer tasks and smaller task sizes than large customers. At the beginning of the sample in May 2016, small customers have 43 tasks per month on average while large customers have 81 tasks. This represents the number of distinct computing projects for a customer in a region-OS per month. Meanwhile, the number of tasks for small customers grows about 4 times as fast as that of the large customers over time, capturing faster growth rates of small firms. By June 2018, at the end of our sample, an average small customer has 89 tasks per month while an average large customer has 92 tasks, suggesting that firm size differences are shrinking on the cloud. Small customers also have on average 71% smaller tasks than large customers.

As a robustness check, we re-estimate the model using only the RDD moments to identify the adoption cost and obtain similar results (Table F14). One might also worry that our specification—modeling inertia only as an adoption cost—misses switching frictions between products the customer has already used. We therefore estimate an extension with switching costs included and find broadly consistent results (Appendix D).

[16]For each customer i, the equivalent quality increases are calculated as $\delta_i/(\beta_i^2 - \beta_i^1)$, averaged across customers.

We do not model switching costs in the baseline for two reasons. First, switching costs in the literature are typically identified under the assumption that each customer chooses one product at a time, so switching events are clearly observed; this is not the case in our setting. Second, continuous usage compounds the problem: when a customer's usage on a product drops, we cannot distinguish a negative demand shock from a switch to another product, which would bias the implied switching-cost estimate.

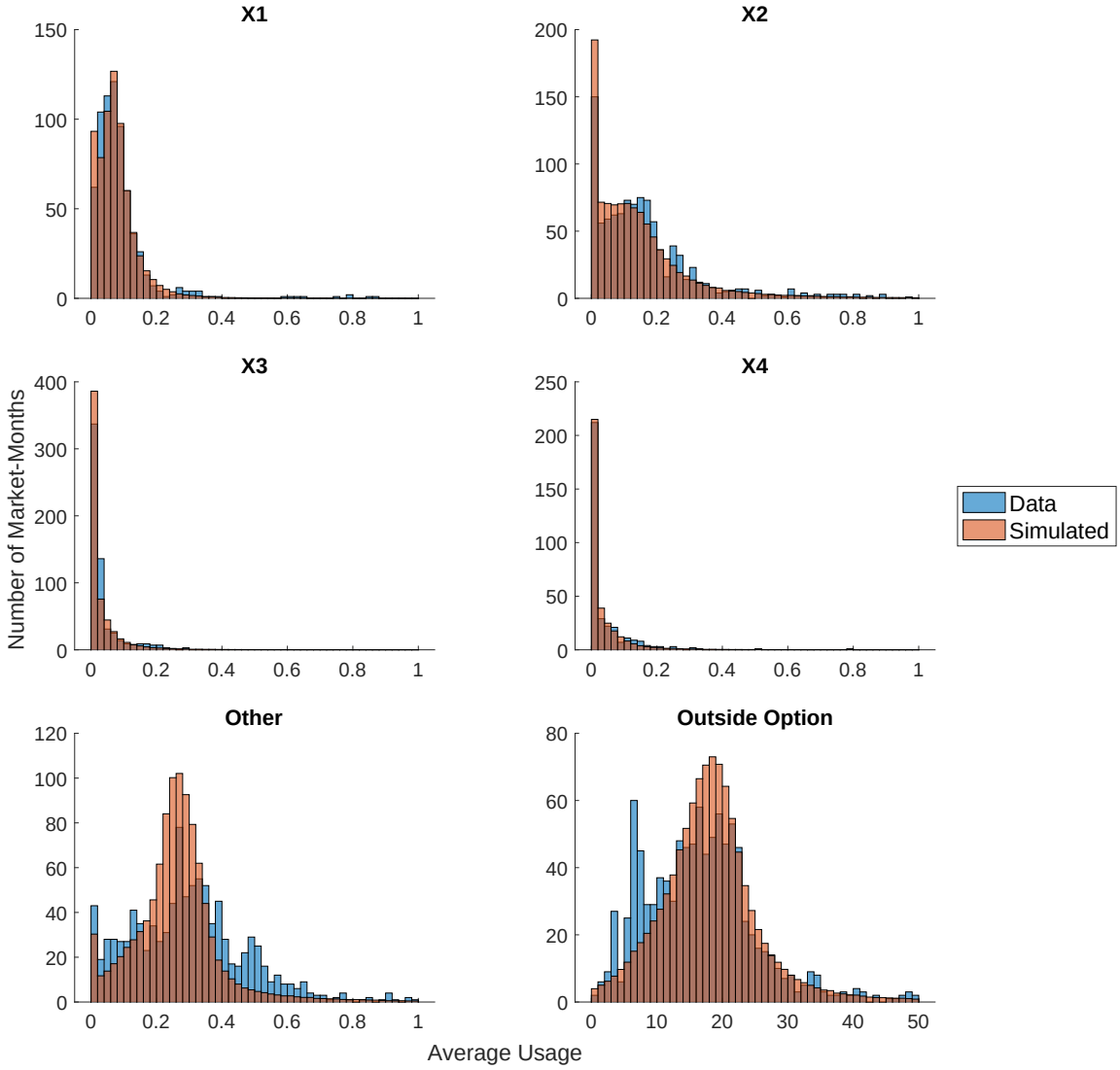
6.1 Model Fit

To assess model fit, we initialize each customer's adoption history with their first-month usage and simulate the model forward over the remaining months. In particular, the new-product indicator New_{ijm} evolves endogenously: updated from simulated rather than observed choices. We then compare simulated usage to the data.

The simulation proceeds as follows: (i) for each customer, take its first month's data as given to initialize which products it has used; (ii) from the second month, for every customer in every market, draw number of tasks and task sizes from the estimated distributions; (iii) for each task, draw i.i.d. type-I extreme value shocks and random coefficients for each product, and simulate product choice based on estimated utilities; (iv) aggregate usage for each customer-product; every time a customer uses a new product, update the new product indicator in its utility function; (v) iterate until the end of the sample.

Figure 4 shows that our estimated model fits the data well. For each simulation, we calculate the distribution of average market-month usage for each product, and then average across 200 simulations. The simulated usage captures well the variation in usage across products and markets. For example, the simulated usage tracks the low usage of X4 in the data despite its high estimated utility, suggesting that our model captures the effect of customer inertia on newer products. The model also predicts well the frequencies of zero usage for each product in the sample. In particular, Figure 5 shows that the model replicates the probabilities of zero usage on the newly launched products for *New* and *Existing* customers who joined the cloud provider just before and just after the launches, providing a good fit of the RDD moments. Finally, Figure F6 and Figure F7 in the Appendix show that the model also fits well in the time series, both in terms of total computing demand and cloud usage by product and customer size.

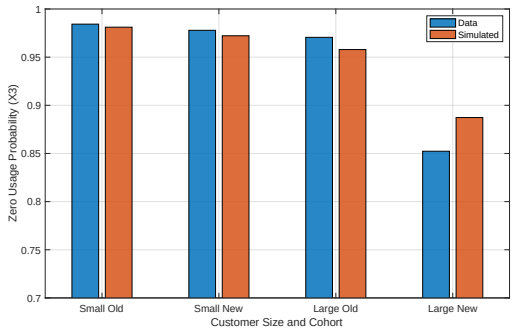
Figure 4: Model fit - Distribution of average market-month usage for each product



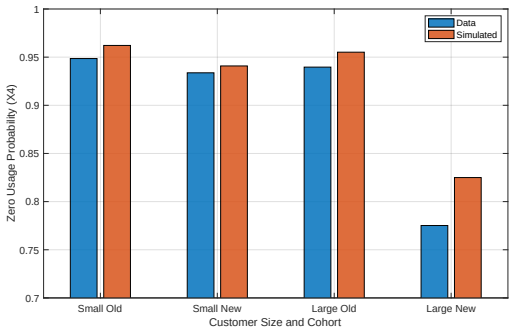
Notes: Figure shows histograms of average usage in a market-month for each product based on data vs. simulations. The blue histograms are based on the data, and the orange histograms are based on simulations. Each bar represents the number of market-months that fall within the corresponding average usage for that product. The simulation results are averaged across 200 simulations.

Figure 5: Model fit - RDD moments

(a) X3 Launch



(b) X4 Launch



Notes: Figure shows the probabilities of having zero usage on the newly launched products for customers of different sizes and cohorts based on data vs. simulations. Panel (a) shows the probabilities of using X3 in a market-month and panel (b) shows the probabilities of using X4 in a market-month. Existing (New) customers are those who joined the cloud provider within the 3 months preceding (following) the launch of the new product. The blue bars are based on the data, and the orange bars are based on simulations. The simulation results are averaged across 200 simulations.

7 Counterfactuals

With our model and estimates, we conduct two sets of counterfactual analyses. First, we quantify the welfare cost of inertia in the public cloud market, by comparing the estimated baseline with inertia to a counterfactual where inertia is eliminated. Second, we study a policy intervention that removes inertia entirely via an "automatic migration service", often offered by cloud providers or third-party specialized vendors. We evaluate the cost of providing this service during our sample period and whether it yields welfare gains that justify private or public provision of this service.

We conduct the counterfactuals under partial equilibrium with the supply side fixed, as in Einav, Klopack and Mahoney (2025). This choice is motivated by the scope of our data and the public cloud market structure. Our data and empirical strategy focus on the within-provider product choice; we do not observe customers' usage of competing providers at the granularity required to model provider choices. On the demand side, this within-provider margin captures most of the variation during our sample period because multi-homing was rare. On the supply side, however, the major cloud providers priced for long-run growth rather than short-run profit maximization, so observed prices do not reflect a static optimum. In the absence of marginal-cost data and a structural model of how the provider sets prices, the supply-side response to removing inertia is not well-identified. The relevant forces work in opposite directions: removing inertia can intensify competition and lower prices, but it can also expand demand for the cloud provider, leading it to charge higher prices. Which force dominates depends on competition structure we cannot characterize

from the current data. A full endogenous-pricing exercise is a natural direction for future work.

7.1 Welfare Cost of Inertia

To quantify the welfare cost of inertia, we compare the estimated baseline (with inertia) to a counterfactual in which inertia is eliminated, holding the supply side fixed. We simulate product choices and usage forward in each scenario using the procedure from Section 6.1, averaging outcomes over 200 simulations per customer. Consumer surplus is computed as the sum of realized task-level utilities, converted to dollars by dividing by the price coefficient; provider revenue is total usage on each product multiplied by its price. We define total welfare as the sum of consumer surplus and provider revenue. We use revenue in place of producer surplus because we do not observe the provider's exact costs; because the marginal cost of serving additional usage on deployed capacity consists largely of electricity and is small relative to price, this is a close approximation. Table 7 reports the welfare estimates. Row 1 is the baseline (with inertia), row 2 is the counterfactual (without), and row 3 reports the difference. Because dollar amounts are re-scaled for confidentiality, we focus on percentage changes.

Table 7: Welfare cost of inertia

		Consumer Surplus (10^3)		**Provider Revenue (10^3)**		**Customer ROI (CS/Revenue)**		**Cloud Usage (10^3)**	
		(Small)	(Large)	(Small)	(Large)	(Small)	(Large)	(Small)	(Large)
Baseline: with inertia	By size	13.2	38.2	5.0	18.9	2.6	2.0	6.7	29.0
		(0.1)	(0.3)	(0.04)	(0.2)	(0.02)	(0.02)	(0.1)	(0.3)
	Total	51.5		23.9		2.2		35.7	
		(0.3)		(0.2)		(0.01)		(0.3)	
Counterfactual: without inertia	By size	33.0	103.3	10.9	45.8	3.0	2.3	17.3	82.0
		(0.2)	(0.5)	(0.1)	(0.2)	(0.02)	(0.01)	(0.1)	(0.4)
	Total	136.3		56.7		2.4		99.2	
		(0.5)		(0.2)		(0.01)		(0.4)	
Δ Welfare (without - with)	By size	19.8	65.1	5.9	26.9	0.4	0.2	10.6	53.0
		(0.1)	(0.4)	(0.05)	(0.2)	(0.01)	(0.01)	(0.1)	(0.4)
	Total	84.8		32.8		0.2		63.6	
		(0.5)		(0.2)		(0.01)		(0.4)	

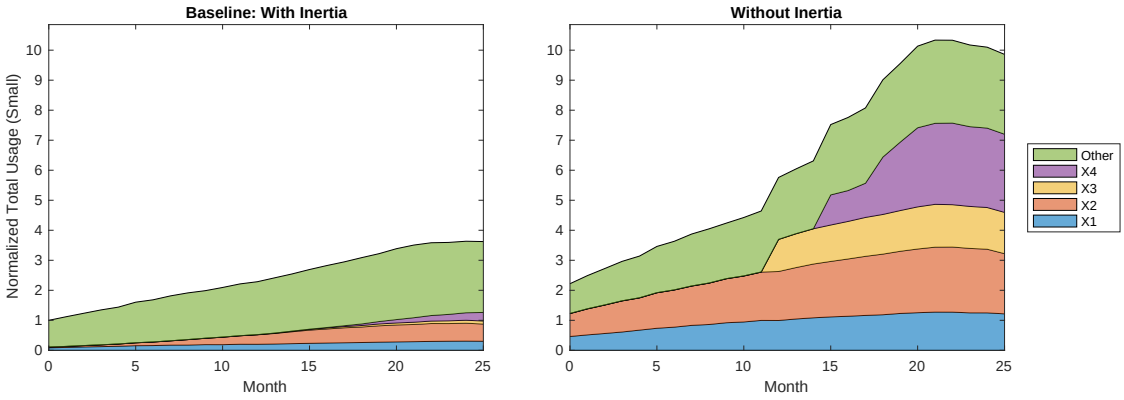
Notes: Table shows consumer surplus, provider revenue, cloud customer ROI, and total cloud usage in the estimated model (baseline, with inertia) and in a counterfactual where inertia is eliminated, with the implied welfare cost given by the difference. Consumer surplus and provider revenue are shown in re-scaled dollars. Customer ROI is calculated as consumer surplus divided by provider revenue, where provider revenue is equal to cloud spending. Cloud usage is shown in re-scaled compute units. Results are based on 200 simulations, with standard deviations in parentheses.

Inertia reduces consumer surplus by 62% relative to the no-inertia counterfactual. The loss has two components. First, inertia distorts product choice within the cloud: customers continue using older, inferior, and more expensive products long after better alternatives become available. Figure 6 decomposes cloud usage by product. Without inertia, usage

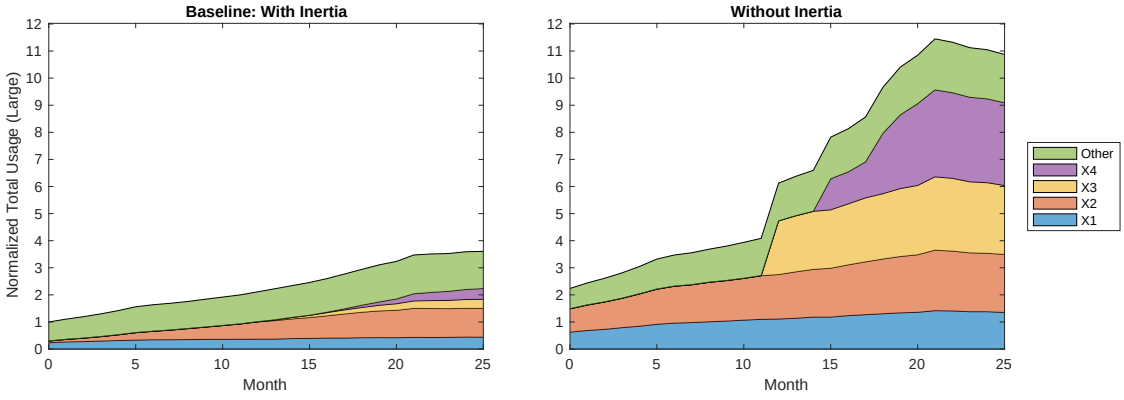
shifts to X3 and X4 within one month of their respective launches; with inertia, usage on these products remains negligible for more than five months. Second, because the cloud as a whole is less attractive when newer products are not adopted, the migration of workloads from private IT onto the provider that would have occurred absent inertia is suppressed. Both components compound over time as inertia persists across successive product launches.

Figure 6: Impact of inertia on usage by product and customer size

(a) Small customers



(b) Large customers



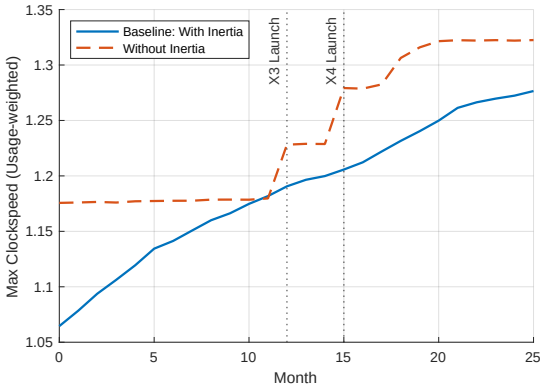
Notes: Figure shows cloud usage by product over time (months since beginning of sample), simulated with adoption cost (left) and without adoption cost (right). Figure 6a shows the usage for small customers and Figure 6b shows the usage for large customers. For each customer size, the usage is normalized with respect to the total cloud usage with adoption cost in month zero.

The consumer surplus loss of 62% is computed by treating the estimated inertia as a dis-

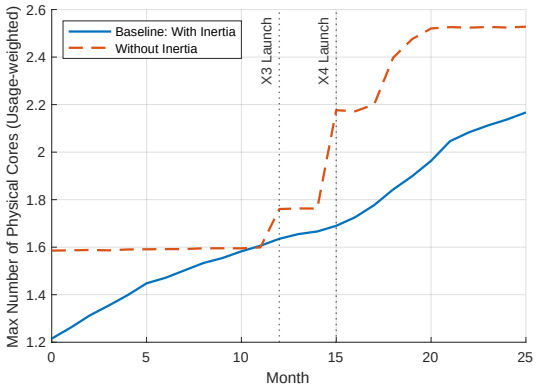
utility, i.e., a real cost paid by the customer. If one instead treats inertia as a non-pecuniary friction that is welfare-irrelevant and excludes it from the baseline consumer surplus, the surplus loss will be smaller by less than 2 percentage points, because it is dominated by misallocation toward inferior products rather than by the adoption cost itself. Table F15 presents a comparison of the consumer surplus under these two treatments of inertia.

Figure 7: Inertia leads to technology misallocation

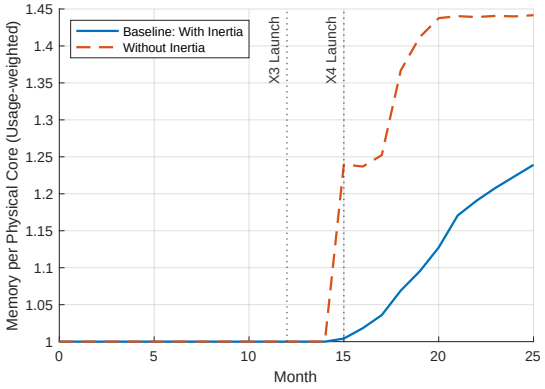
(a) Clockspeed



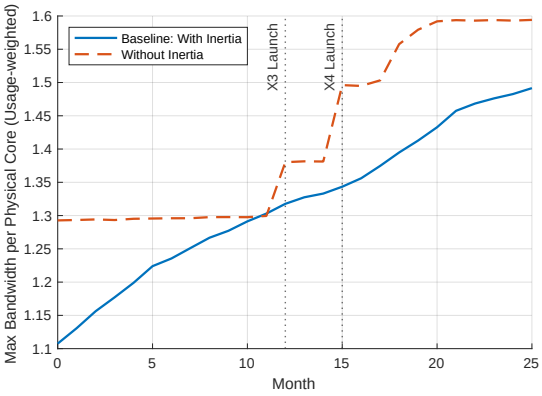
(b) Number of physical cores



(c) Memory



(d) Bandwidth



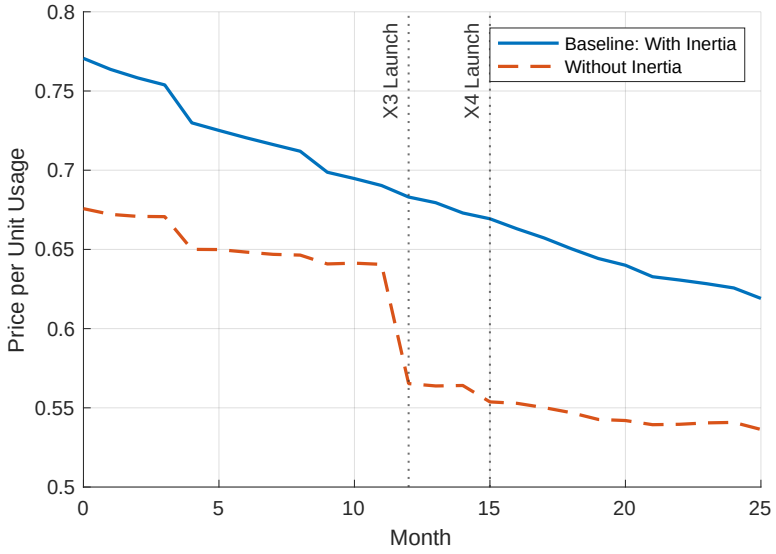
Notes: Figure plots the average *Max Clockspeed* (Figure 7a), *Available Number of Physical Cores* (Figure 7b), *Memory per Physical Core* (Figure 7c), and *Bandwidth per Physical Core*(Figure 7d) weighted by usage in the X-series, with and without inertia. These product characteristics are presented as ratios relative to X1 as in Table 1. The vertical dotted lines mark the months when X3 and X4 are launched.

Inertia slows the rate at which cloud customers adopt newer technology, leaving them on lower-quality vintages for longer. Figure 7 plots the usage-weighted average product quality within the X-series, with and without inertia, along four dimensions salient for cloud customers: clockspeed, physical cores, memory per core, and bandwidth per core. The X-series has improved substantially along each: relative to X1, X4 offers roughly 50% higher clockspeed, four times as many physical cores, more than twice the memory per

core, and twice the bandwidth per core (Table 1). With inertia, the cloud products customers actually use are on average 4% slower in clockspeed, with 14% fewer cores, 9% less memory per core, and 7% lower bandwidth per core than in the no-inertia counterfactual. The two paths converge just before X3's launch, roughly two years after X2 was introduced—by which point inert customers have caught up to the no-inertia level on X2. The welfare-relevant point is timing: absent inertia, customers would have captured these productivity gains years earlier.

Inertia also raises the average price customers pay per compute unit. On average, cloud customers pay 17% ($= \frac{23.9/35.7}{56.7/99.2} - 1$) more per unit of cloud usage due to inertia (Table 7).[17] Figure 8 plots this price with and without inertia. Even with inertia, unit prices fall by 20% over the sample period as technology advances and cheaper new products become available. Eliminating inertia would lower the average price by an additional 13%, reflecting the gain customers forgo by failing to adopt new products immediately upon launch.

Figure 8: Inertia increases the unit price of cloud usage



Notes: Figure plots the usage-weighted unit price of cloud computing with and without inertia, expressed in re-scaled dollars per compute unit. The vertical dotted lines mark the months when X3 and X4 are launched.

Due to this misallocation, inertia makes the public cloud less attractive overall. This leads to a 64% reduction in total cloud usage from the provider relative to the no-inertia

[17] Corresponding statistics are 45% in Handel (2013) for health insurance, 21–28% (median) in Dubé, Hitsch and Rossi (2010) for consumer goods, and 18% in Brown et al. (2023) for institutional index funds. Our magnitudes are thus largely consistent with the literature and plausible in an enterprise IT procurement setting.

counterfactual. By the end of the sample period, no-inertia cloud usage is roughly three times the baseline for both small and large customers (Figure 6).

As a result, provider revenue is 58% lower in the presence of inertia, even though customers locked in on older products pay more per unit of computing. This might seem surprising in light of the classical switching-cost intuition (Klemperer, 1995; Farrell and Klemperer, 2007), under which provider rents should rise when customers are locked in. That mechanism is still present in our data—customers stuck on X1 pay 23% more per compute unit than they would on X4—but it is dominated by a second channel: newer products expand cloud demand for the provider, and inertia keeps customers on inferior vintages that compete poorly against the outside option. The provider's loss from this lowered demand volume far exceeds the additional per-unit rents it extracts from existing customers.

Despite the inertia, the public cloud generates substantial welfare gains. The average customer ROI—consumer surplus divided by provider revenue, where provider revenue is equal to cloud spending—is 2.2x (Table 7). Assuming this ratio holds across the broader IaaS market, which generated about $26 billion in annual revenue during our sample period, the implied annual consumer surplus from IaaS exceeds $50 billion. Small customers have higher ROIs (2.6x) than large customers (2.0x) even though their total surplus is smaller, consistent with cloud usage disproportionately benefiting small firms by allowing them to scale computing needs and access newer technology. Eliminating inertia would raise the average ROI by 9%, from 2.2x to 2.4x.

Our estimated ROI is comparable to estimates for other computing technologies: higher than that of PCs (0.92x; Hendel, 1999) but lower than that of mainframes (3.3–6.1x; Bresnahan, 1986). Following the prior literature, we interpret it as an upper bound because it reflects benefits to existing cloud customers without netting the costs of transitioning to the cloud. The estimate also reflects pricing in a still-nascent industry: cloud providers in our sample period prioritized long-run growth over short-run profit (Castillo, 2025), so early adopters may extract more surplus than they will once the industry matures.

7.2 Automatic Migration

Cloud providers and specialized third-party vendors already offer migration assistance ranging from informational consulting to hands-on, end-to-end migration projects (e.g., AWS's Migration Acceleration Program, Azure's Migration and Modernization Center, and vendors such as Cloudreach and Rackspace). We model a counterfactual in which such a service is universally available, fully eliminating the adoption cost for any customer who

tries a new product. We implement this by subsidizing adoption costs whenever a customer adopts a new product, so customers face the same effective choice problem as in the no-inertia counterfactual of Table 7.

Cloud migrations in general can involve data sovereignty considerations, configuration-specific dependencies, testing requirements, downtime risk, etc. Our setting attenuates these issues by construction: because the migration we consider stays within the same provider, same data center, and same product series, data sovereignty is unchanged, configurations transfer essentially without modification, and testing and downtime requirements are minimal. Providers and third-party vendors routinely execute cross-provider migrations that are technically more demanding than the within-series migrations we consider. What we model as "automatic" is best understood as the provider or vendor absorbing the organizational friction—through personalized notification, pre-configured tooling, free testing credits, or hands-on customer-success engagement—for customers who would otherwise default to their existing configuration.

Table 8: Cost of full cloud migration service (10^3)

Product	All Customers	Small	Large
X1	10.0	2.5	7.5
	(0.2)	(0.1)	(0.2)
X2	10.8	3.1	7.7
	(0.2)	(0.1)	(0.2)
X3	15.2	3.0	12.2
	(0.4)	(0.1)	(0.4)
X4	18.2	4.3	13.9
	(0.4)	(0.1)	(0.4)
Other	4.6	0.8	3.7
	(0.1)	(0.0)	(0.1)
Total	58.7	13.6	45.1
	(0.7)	(0.2)	(0.7)

Notes: Table presents the cost of full cloud migration service in re-scaled dollars during our sample, broken down by products and customer sizes. Results are averaged across 200 simulations, with standard deviations in parentheses.

Table 8 reports the cost of implementing this policy in re-scaled dollars, broken down by product and customer size. We compute it by simulating customer choices under the policy and summing the adoption costs incurred whenever customers adopt a product they have never used before. The reported numbers should be read as upper bounds: they treat the full estimated adoption cost as a real resource cost that some provider must absorb. Because a provider can reuse migration tooling across many customers, and to the extent

the estimated inertia reflects organizational friction rather than a real resource cost, the actual resource cost of automating the migration is smaller. Even at this conservative upper bound, the total cost across all customers and products is at most 58.7k, while the policy raises total welfare by 117.6k over the sample period (= 84.8 + 32.8, from Table 7). The policy therefore generates at least two dollars of welfare for every dollar spent.

Who would fund such a policy? The cloud provider itself is one possibility. Over our sample period, the provider's additional revenue from eliminating inertia (32.8k) is below the upper-bound migration cost (58.7k), but the migration cost is typically incurred early on while revenue benefits accrue over time, so in the long run the provider may recoup the investment. Appendix E shows that over longer horizons the cloud provider can benefit from offering automatic migrations, as well as introductory discounts for new products, and that such policies Pareto-improve both provider revenue and consumer surplus. Public financing is a second possibility. Existing public programs already subsidize digital adoption and broadband upgrades;[18] in our setting, the total welfare gain from funding cloud migrations exceeds even the upper-bound cost by a factor of two, so public support may be justifiable as cloud computing becomes more ubiquitous.

8 Conclusion

In principle, the public cloud allows firms to adopt newer computing technology at the click of a button. We document substantial customer inertia even in a "best-case" switching environment—within the same provider, data center, and product series. A regression discontinuity around new-product launches shows that customers entering just after a launch are 15 percentage points more likely to adopt the new product than otherwise similar customers entering just before, indicating that the friction is driven primarily by organizational decision-making rather than by technical barriers or migration fees.

We develop and estimate a structural demand model that jointly captures multiple product choice and continuous usage to quantify the welfare impact of inertia on cloud. We find that this inertia destroys 62% of consumer surplus and 58% of provider revenue. It slows the diffusion of productivity-enhancing hardware to firms and raises the unit price they pay for computing. The provider-revenue result runs against the classical switching-cost intuition under which consumer inertia benefits incumbents through market power. In our setting, inertia keeps customers on older products that compete poorly against private

[18] See https://www.imda.gov.sg/how-we-can-help/smes-go-digital and https://www.gov.uk/government/publications/gigabit-broadband-voucher-scheme-information/gigabit-broadband-voucher-scheme-information, accessed Jul. 23rd 2025.

IT, so the provider loses the demand volume it would have captured if the inertia were absent.

Despite the inertia, the public cloud still delivers large returns to customers with a 2.2x ROI on average. This is comparable to major twentieth-century IT inventions such as mainframes and PCs, with over $50 billion in implied annual consumer surplus during our sample period. It also disproportionately benefits small firms (Bloom and Pierri, 2018). Policies that absorb the organizational friction could capture even more of this welfare: a migration service would generate at least two dollars of total welfare for every dollar spent, and Pareto-improve consumer surplus and provider revenue over long horizons.

We view this paper as a first step toward understanding the public cloud market. Our analysis abstracts from choice between providers, forward-looking customers, and supply-side responses, which are promising directions for future research. Our methodology and mechanism may extend beyond cloud VMs to other usage-based digital input markets in which firms allocate continuous usage across multiple products with heterogeneous prices and quality. A natural near-term application is enterprise adoption of large language models, where firms route tasks across multiple model tiers and generations offered by the same provider under per-token pricing, and where the same set-it-and-forget-it organizational friction we document plausibly slows technology turnover.

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Online Appendix

A Proofs

A.1 Challenge of Maximum Likelihood Estimation

In this section, we discuss challenges of maximum likelihood estimation for our model. Suppose for now that task sizes are known and equal to one, i.e., $q_{im} = 1$. Then our model collapses to that of Burda, Harding and Hausman (2012), and the likelihood of $(y_{i0m}, y_{i1m}, ..., y_{iJm})$ can be written as

$$\begin{aligned} &L_i(\lambda_{im}, \alpha_i, \beta_i, \delta_i | y_{i0m}, y_{i1m}, \cdots y_{iJm}) \\ =&\frac{\exp(-\lambda_{im})\lambda_{im}^{\sum_{j=0}^{J} y_{ijm}}}{(\sum_{j=0}^{J} y_{ijm})!} \cdot \prod_{j=0}^{J} P_{ijm}(1)^{y_{ijm}}, \end{aligned}$$

which is a product of two components: 1) the probability of receiving $\sum_{j=0}^{J} y_{ijm}$ tasks; and 2) the probability that VM j is chosen for exactly y_{ijm} tasks, for all j. This likelihood is easy to compute because the task size is known. However, if the task size is unknown, the likelihood becomes the sum of the probabilities of all scenarios that generate usages $(y_{i0m}, y_{i1m}, ..., y_{iJm})$. For example, if we observe usage of $y_{ijm} = 10$, it could be generated by one task of size 10, two tasks of sizes $(1, 9)$, $(2, 8)$, and so on, amounting to infinite possibilities with continuous usage. In other words, the likelihood becomes an infinite sum of convoluted integration, which is computationally infeasible. Therefore, we take the GMM approach instead.

A.2 Proof of Proposition 1

Proof. For the simplified model without inertia, customer heterogeneity, or time trend in the number of tasks, let the probability of customer i choosing product j in market m for a task of size q_{im} be denoted by $P_{ijm}(q_{im})$, i.e.,

$$P_{ijm}(q_{im}) = \begin{cases} \frac{\exp(\alpha p_{jm} q_{im} + X_j \beta)}{1 + \sum_{l=1}^{J} \exp(\alpha p_{lm} q_{im} + X_l \beta)} & \text{if } j \neq 0, \\ \frac{1}{1 + \sum_{l=1}^{J} \exp(\alpha p_{lm} q_{im} + X_l \beta)} & \text{otherwise.} \end{cases}$$

Zero Usage Probability Moment For any given number of tasks n_{im} and task size q_{im}, the probability of customer i not using product j in market m is equal to the probability of cus-

tomer i not choosing product j in market m for any of the n_{im} tasks. Because tasks are independent, this probability is equal to $(1 - P_{ijm}(q_{im}))^{n_{im}}$, which we call $Pr(\text{Product Choice})$. Now, to obtain the probability of zero usage at the customer-product level, we first integrate the probability of product choice over the exponential distribution of task sizes, and then the Poisson distribution of the number of tasks:

$$\mathbb{P}(y_{ijm} = 0) = \sum_{n_{im}=0}^{\infty} \underbrace{\frac{\exp(-\lambda)\lambda^{n_{im}}}{n_{im}!}}_{Pr(\text{Number of Tasks})} \cdot \int_{q_{im}} \underbrace{(1 - P_{ijm}(q_{im})\ )^{n_{im}}}_{Pr(\text{Product Choice})} \underbrace{\frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im})}_{Pr(\text{Task Size})} dq_{im}.$$

Expected Usage Moment Again, first fix the number of tasks to be n_{im}. By integrating over the exponential distribution of task sizes and the probability of product j being chosen, we obtain the average task size on product j:

$$\int_{q_{im}} q_{im} P_{ijm}(q_{im}) \frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im}) dq_{im}.$$

Then, we multiply it by n_{im} to get the expected usage for each product given n_{im} tasks, because task sizes are the same across tasks for the same customer in the same market. Finally, integrating over the Poisson distribution of the number of tasks, we obtain the expected usage at the customer-product level:

$$\begin{aligned}
\mathbb{E}(y_{ijm}) &= \sum_{n_{im}=0}^{\infty} \frac{\exp(-\lambda)\lambda^{n_{im}}}{n_{im}!} \cdot \left(n_{im} \int_{q_{im}} q_{im} P_{ijm}(q_{im}) \frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im}) dq_{im} \right) \\
&= \left(\sum_{n_{im}=0}^{\infty} \frac{\exp(-\lambda)\lambda^{n_{im}}}{n_{im}!} \cdot n_{im} \right) \int_{q_{im}} q_{im} P_{ijm}(q_{im}) \frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im}) dq_{im} \\
&= \underbrace{\lambda}_{\mathbb{E}(\text{Number of Tasks})} \underbrace{\int_{q_{im}} q_{im} P_{ijm}(q_{im}) \frac{1}{\gamma}\exp(-\frac{1}{\gamma}q_{im})\, dq_{im}}_{\mathbb{E}(\text{Task Size})},
\end{aligned}$$

where λ is exactly the expected number of tasks of the Poisson distribution.

□

A.3 Proof of Own-Price Elasticity

In this section, we derive the formula for the own-price elasticity, $\frac{\partial \mathbb{E}(y_{ijm})}{\partial p_{jm}} \frac{p_{jm}}{\mathbb{E}(y_{ijm})}$. Following Proposition 1, the expected usage of customer i on VM j in market m is

$$\mathbb{E}(y_{ijm}) = \lambda_{im} \int_{q_{im}} q_{im} P_{ijm}(q_{im}) \cdot \frac{1}{\gamma_i} \exp(-\frac{1}{\gamma_i} q_{im}) dq_{im}.$$

Differentiating the expected usage with respect to its price:

$$\frac{\partial \mathbb{E}(y_{ijm})}{\partial p_{jm}} = \lambda_{im} \int_{q_{im}} \alpha_i q_{im}^2 \cdot P_{ijm}(q_{im})(1 - P_{ijm}(q_{im})) \frac{1}{\gamma_i} \exp(-\frac{1}{\gamma_i} q_{im}) dq_{im}.$$

Finally, the own-price elasticity is given by

$$\frac{\partial \mathbb{E}(y_{ijm})}{\partial p_{jm}} \frac{p_{jm}}{\mathbb{E}(y_{ijm})} = \frac{\alpha_i p_{jm} \int_{q_{im}} q_{im}^2 \cdot P_{ijm}(q_{im})(1 - P_{ijm}(q_{im})) \exp(-\frac{1}{\gamma_i} q_{im}) dq_{im}}{\int_{q_{im}} q_{im} P_{ijm}(q_{im}) \exp(-\frac{1}{\gamma_i} q_{im}) dq_{im}}.$$

B Data and Estimation Details

B.1 Prices, Instruments, and First Stage Results

We first construct a price index for the prices of the "other" VM across markets, which include all of the provider's VM products other than the X series. Specifically, in each market, we calculate the price index as the average price weighted by the demand of each product from 2017, so that the variation of the price index over time is driven by changes in product availability and price changes, rather than demand.

We use three cost shifters as instruments for prices. Cloud providers' variable costs vary across regions and products due to regional differences in electricity costs and because the same VMs are supported on different hardware across regions. Our first instrument uses 2019 electricity prices for each region to capture the provider's regional variation in costs.

The second instrument uses power ratings of different types of hardware and the third instrument uses costs of procuring an additional cluster of different types of hardware. Because different VMs can run on different hardware and the allocation of VMs onto different hardware clusters differ across regions, we transform the hardware costs and power ratings at the hardware type level to be at the VM level, weighting them by demand from 2017. As a result, both the power rating and the hardware cost instruments vary at the region-product level.

The first stage results of the control function are shown in Table B1. We regress prices

Table B1: Control function first stage results

	VM Price
Electricity price	0.092
	(0.015)
VM (hardware) cost	1.151
	(0.065)
VM (hardware) power rating	0.766
	(0.078)
X1	0.195
	(0.025)
X2/X3	0.136
	(0.027)
X4	-0.050
	(0.018)
Promo	-0.112
	(0.007)
Constant	-0.933
	(0.106)
Adj. R^2	0.600
Num. obs.	3790

Notes: Table presents results from the first stage in the control function approach. We regress VM prices at the product-market level on the three cost shifters as well as product characteristics. Standard errors are in parentheses.

of all cloud products (X1–X4 and Other) at the product-market level on the three cost shifters, as well as the product characteristics X_j. The estimated coefficients of the cost shifters are statistically significant and have the expected signs: VM prices increase with electricity prices and VM hardware costs. VM prices also increase with VM hardware power rating, i.e., VMs with higher energy consumption (higher rating) are more expensive.

B.2 Computation Details

To calculate the GMM objective function, we numerically integrate over the distributions of task sizes, random coefficients, as well as the number of tasks. For task sizes, we use the Gauss-Laguerre quadrature with 10 nodes. For the random coefficients, we use the Halton sequence with 20 nodes. For the number of tasks, we compute moments for any k number of tasks, for $k = 0, 1, 2, ..., 120$. For large customers at the end of our sample (i.e., largest number of tasks), 120 tasks cover 99.8% of the distribution of the number of tasks.

To speed up the computations, we parallelize $k = 0, 1, 2, ..., 120$ across 28 cores. In total, this integration procedure yields $10 \times 20 \times 121/28 = 864$ points to compute per core. On a 2.6 GHz Intel Gold 6132 CPU with 28 cores, one evaluation of the objective function with gradient takes about 162 seconds, using 508G of memory. We are thus limited by the curse of dimensionality to further increase the number of nodes for the numerical integrations. Finally, we minimize the GMM objective function using the BFGS quasi-Newton algorithm supplied with the analytical gradient.

C Monte Carlo Simulations

We conduct Monte Carlo simulations to illustrate the identification of our model and the finite-sample performance of our estimation procedure.

The data generating process is as follows. We consider the simplified model, which does not have inertia, customer heterogeneity, or time trend in the number of tasks. We parameterize the Poisson distribution for the number of tasks by its mean λ, and the exponential distribution for the task size by its mean γ. We consider markets each with four products and an outside option. Customer i's utility from choosing product j in market m for task k is given by

$$u_{ijmk} = \alpha p_{jm} q_{im} + X_j \beta + \epsilon_{ijmk},$$

where X_j's are dummies for the four products, and the idiosyncratic shocks ϵ_{ijmk}'s are

distributed type-I extreme value. We normalize the mean utility for the outside option to zero.

We fix the parameters at some true values and simulate two samples, both with 100,000 customers per market, but one with 100 markets and the other with 500 markets (see Table C2). We draw i.i.d. prices from a normal distribution at the product-market level. Then, we draw each customer's number of tasks and task size, and simulate their product choices and usage based on their utility function and the prices.

Then, we estimate the model using our GMM estimation procedure in these samples and present the results in Table C2. We recover estimates that are close to the true parameters and their biases decrease slightly with more markets (and thus more price variation).

Table C2: Monte Carlo simulation results

	True Value	Initial Value	Sample 1	Sample 2
No. Customers			100k	100k
No. Markets			100	500
Product choice:				
Price $\times$ task size	-10.000	-15.000	-9.995	-9.998
Product 1 FE	1.000	1.000	1.002	1.001
Product 2 FE	1.500	1.000	1.503	1.500
Product 3 FE	2.000	1.000	2.001	2.000
Product 4 FE	2.500	1.000	2.501	2.501
Number of tasks:				
Mean (λ)	2.000	1.000	1.999	2.000
Task size:				
Mean (γ)	0.300	-2.303	0.300	0.300
Price elasticity:				
Price Elasticity			1.724	1.797
Runtime (s)			2575	13080

Notes: Table presents results from Monte Carlo simulations. We generate two samples, both with 100,000 customers per market, but one with 100 markets and the other with 500 markets. Both samples are generated based on true values of the parameters. Prices vary at the product-market level and are drawn from an i.i.d. normal distribution. The estimation follows our GMM procedure and starts at the initial values of the parameters.

As a comparison, we also fit the standard logit model in the first sample. As logit models do not explain usage, we fix the total usage for each customer-market at the sample values.

Then, we estimate two utility specifications:

$$u_{ijmk} = \alpha p_{jm} + X_j\beta + \epsilon_{ijmk}, \tag{7}$$

$$u_{ijmk} = \alpha_0 p_{jm} + \alpha_1 p_{jm} Q_{im} + \alpha_2 Q_{im} + X_j\beta + \epsilon_{ijmk}, \tag{8}$$

where Q_{im} is the total usage for customer i in market m. By including Q_{im} in the second specification, we attempt to control for the impact of usage on choices.

Table C3: Logit estimates in Sample 1

	True Value	Logit (1)	Logit (2)
Price elasticity	1.724	7.224	7.187

Notes: Table presents results from estimating logit models in Sample 1 of Table C2. Logit (1) uses utility specification in equation (7). Logit (2) uses utility specification in equation (8).

Table C3 presents the true average price elasticity and the average price elasticities implied by the estimated logit models. The elasticities implied by the logit models are severely biased upwards. Intuitively, this is because consumers are naturally more sensitive to prices when they know that they will incur high usage. They tend to choose cheaper products when the task size is large and care less about prices when the task size is small. Logit models mistake this large usage on the cheap products as sensitivities to prices that are independent from usage. This comparison highlights the importance of our structural model.

D Extension: Switching Cost

In this section, we estimate an extension of the model where we allow switching cost to affect consumer demand in addition to adoption cost. Formally, the customer's utility from using VM j in market m for task k is now:

$$u_{ijmk} = \alpha_i p_{jm} q_{im} + X_j\beta_i + \delta_i^{adopt} \textit{New}_{ijm} + \delta_i^{switch} \textit{Switch}_{ijm} + \xi_{ijm} + \epsilon_{ijmk}, \tag{9}$$

where $\textit{Switch}_{ijm}$ is a binary variable indicating that customer i has used VM j in the past but switched away from it before market m. The switching cost is represented by δ_i^{switch}, which varies by customer size.

Because a customer can use multiple products at the same time in our setting, this switching cost δ_i^{switch} is different from what is typically estimated in the literature (MacKay

and Remer, 2024). This is the cost of "restarting" usage on a previously used VM j regardless of what other VMs customer i is using, rather than the cost of switching from some VM k to VM j ($k \neq j$).

In practice, we measure $\textit{Switch}_{ijm}$ with a binary variable that is equal to 1 if and only if customer i has used VM j in the past but did not use VM j last month. In other words, denoting the market last month by m', we proxy $\textit{Switch}_{ijm}$ with $(!\textit{New}_{ijm})\,\&\,(y_{ijm'} = 0)$. This proxy of $\textit{Switch}_{ijm}$, however, is subject to measurement error. When we observe that a customer incurs zero usage on product j in a period ($y_{ijm'} = 0$), because the customer faces uncertainty in total demand, we are not able to distinguish between whether the customer is switching away from product j ($\textit{Switch}_{ijm} = 1$), or is simply pausing the usage on product j due to low total demand (e.g., few computation tasks). By proxying $\textit{Switch}_{ijm} = 1$ with $(!\textit{New}_{ijm})\,\&\,(y_{ijm'} = 0)$, we might mistake total demand fluctuations as switching, which could bias the demand estimates.

Table D4: Moment conditions with switching cost

Moments	Characteristics conditioned on	# Moment conditions
Zero Usage Probability	- region, choice set	178
	- $\textit{New}_{ijm}$ (for $j \neq 0$), $y_{ijm'} = 0$ (for $j \neq 0$), customer size	30
	- RDD: $\textit{NewCustomer}_{ij}$, customer size	8
Expected Usage	- region, choice set	222
	- $\textit{New}_{ijm}$ (for $j \neq 0$), $y_{ijm'} = 0$ (for $j \neq 0$), customer size	30
	- month (t_m), customer size	52

With these caveats in mind, we estimate the extended model using the GMM procedure introduced in Section 5. In addition to the moments described in Section 5.1, we also condition the Zero Usage Probability $\mathbb{P}(y_{ijm} = 0)$ and the Expected Usage $\mathbb{E}(y_{ijm})$ on whether the customer used the product last month ($y_{ijm'} = 0$) to identify the switching costs. Table D4 lists all the moment conditions we use to estimate this extension.

Table D5 presents the results. The demand estimates are broadly consistent with the baseline. We estimate negative and statistically significant switching costs for both small and large cloud customers. Compared to the baseline, however, the standard deviation of the random coefficients are also smaller. This suggests the measurement error in $\textit{Switch}_{ijm}$ might have led the extended model to mistake demand uncertainty for switching cost, over-estimating the magnitude of the switching costs (i.e., make them too negative) and under-estimating the demand uncertainty. Nevertheless, the estimated switching costs are much smaller in magnitude—by at least 50%—than the adoption costs, so the adoption costs remain the primary concern in this market.

Table D5: Robustness: Demand estimates with switching cost

Parameter	Estimate	SE	Estimate	SE
Panel (a) Product choice: means	Small		Large	
X1 (β_i^1)	-3.593	(0.173)	-3.355	(0.258)
X2/X3 (β_i^2)	-3.346	(0.119)	-3.260	(0.184)
X4 (β_i^4)	-3.649	(0.124)	-3.166	(0.155)
Other (β_i^O)	-2.668	(2.236)	-3.044	(2.214)
Promo (β_i^P)	-0.870	(0.099)	-0.039	(0.063)
Price $\times$ task size (α_i)	-7.478	(1.721)	-1.833	(0.656)
Adoption cost (δ_i^{adopt})	-5.988	(0.168)	-9.825	(10.228)
Switching cost (δ_i^{switch})	-2.974	(0.073)	-3.694	(0.101)
Control function (κ_i)	1.273	(0.439)	-0.164	(0.314)
Panel (b) Product choice: standard deviations				
X1 (σ^1)			0.030	(0.910)
X2 (σ^2)			0.041	(0.503)
X3 (σ^3)			0.143	(0.260)
X4 (σ^4)			0.753	(0.087)
Other (σ^O)			0.002	(13.326)
Panel (c) Number of tasks	Small		Large	
Intercept (λ_{i0})	47.440	(2.156)	87.530	(3.081)
Time trend (λ_{i1})	2.031	(0.102)	0.498	(0.140)
Panel (d) Task size	Small		Large	
Exponential scale (γ_i)	0.096	(0.003)	0.342	(0.005)

Notes: Table presents estimates of demand parameters. Standard errors are in parentheses and computed using estimated GMM asymptotic variance. We allow all parameters to depend on whether a customer is small or large, except for the standard deviations for the random coefficients in Panel (b). In Panel (a), (c), and (d), the first (last) two columns present the parameter estimates and standard errors for small (large) customers.

E Long-run Policy: Automatic Migration and Introductory Discount

Besides automatic migration services, another common approach to reduce customer inertia is introductory discounts, for new products or new users. In this section, we consider the long-run welfare impact of these policies.

We start with introductory discounts. Like the provider-funded automatic migration service, introductory discounts impose short-run costs on the provider and increases long-run revenue. In addition, the introductory discount also allows the provider to optimally choose the discount percentage and target the right customers to maximize revenue.

We consider two types of introductory discounts. The first is a product-level discount—often referred to as new product preview or "open beta"—which offers a temporary price reduction on a new product following its launch. The second is a customer-product level discount—often referred to as personalized product trial—which discounts a product for any customer who has not used it before. We assume that both types of discounts last for one month, though this assumption can be easily relaxed.

Our analysis focuses on the launch of the new product X4. To isolate the effect of the discounts, we simulate the demand forward from the launch date of X4, holding market characteristics (customers, choice sets, and prices) fixed and only allowing the number of tasks to grow following the time trends. For each type of discount, we first solve for the provider-optimal discount percentage that maximizes the provider's long-run revenue. This allows us to assess whether any discount is profitable for the provider. Then, we evaluate the impact of these provider-optimal discounts on consumer surplus and total welfare. Finally, we explore how targeting discount percentages based on customer size affects the outcomes.

To compute the provider-optimal discounts, we conduct a grid search over a wide range of discount percentages. We assume a monthly discount factor of 0.99 for the provider and average revenues across 1,000 simulations. As the provider trades off short-run costs of discounts vs. long-run revenue benefits, the provider-optimal discounts depend on how far out we simulate. In Appendix E.1, we show that the optimal discounts increase with the provider's planning horizon and stabilize after 96 months, which we use as the planning horizon for our main results.

Table E6 presents the effects of the different remedies relative to the baseline where there are no discounts. We find that the provider-optimal discount for new product preview is 250% (Table E6, Row 1). That is, the provider pays back to the customers, possibly in

Table E6: Effect of different remedies for X4 launch in comparison to the baseline

		% Difference from baseline			
	Provider-optimal discount	Provider revenue	Consumer surplus	Total welfare	Cloud usage
1. New product preview	250%	1.3%	1.9%	1.7%	1.9%
2. New product preview (by customer size)	Small: 325% Large: 250%	1.3%	2.1%	1.8%	2.0%
3. Personalized product trial	225%	13.1%	16.7%	15.7%	17.2%
4. Personalized product trial (by customer size)	Small: 275% Large: 200%	13.4%	16.1%	15.3%	16.1%
5. Provider-funded automatic migration service on X4		12.2%	24.8%	21.1%	21.9%

Notes: Table compares welfare and cloud usage in the baseline (i.e., no discount), product-level introductory discounts (i.e., new product preview), customer-product level introductory discounts (i.e., personalized product trial), and provider-funded automatic migration service on X4. Provider-optimal discounts are calculated based on grid searches over a wide range of discounts and 96 months of simulation after X4's initial launch. Provider revenue, consumer surplus, total welfare, and cloud usage are shown in percentage differences relative to the baseline. All results are averages across 1000 simulations.

store credits, 1.5 times their spending on X4 in the first month post-launch.[19] This discount percentage is high because the customers have low price elasticity but high adoption cost, so the provider needs to pay them a lot to shift their choices. The provider-optimal new product preview increases the provider's revenue by 1.3% and consumer surplus by 1.9%, improving welfare by 1.7% overall, so this is a Pareto improvement. The total cloud usage goes up by 1.9%.

Turning to personalized product trials, we find that the provider-optimal discount is 225%, slightly lower than that of the new product preview (Table E6, Row 3). In other words, the provider should pay customers who have not used X4 an amount that is equal to 1.25 times their first month's spending on X4. Despite its slightly lower discount percentage, the optimal personalized product trial leads to a significant 13.1% revenue increase relative to the baseline. Consumer surplus also increases by 16.7%, total welfare by 15.7%, and total usage by 17.2%. Therefore, the personalized product trial dominates the new product preview both in terms of consumer surplus and provider revenue, as it is not restricted to the first month of the product launch, but is offered whenever a customer tries a new product for the first time.

Allowing the provider to target different customer sizes with different discount per-

[19] Another interpretation of the discount percentage is that it represents the equivalent discount in one month from a multi-month discount campaign.

centages leads to moderate improvement in provider revenue (Table E6, Row 2 and 4). The provider-optimal discounts for new product preview are 325% for small customers and 250% for large customers; for personalized product trial, the optimal percentages are 275% for small and 200% for large. Contrary to what cloud providers typically do in practice, the provider should give a higher discount to smaller customers in our setting. This is because, first, small customers are less price-elastic due to lower usage and thus need a higher discount to overcome similar adoption costs. Second, because small customers have lower usage in the beginning but grow faster, it is cheaper for the provider to discount small customers upfront to encourage new product adoptions, and receive revenue benefits from their higher usage later.

Finally, we compare these introductory discounts with a provider-funded automatic migration service on X4 that fully eliminates the adoption cost for X4 (Table E6, Row 5). Here, we consider its long-run effect by simulating the demand forward over the 96 months following X4's launch, rather than limiting to our sample period as in Section 7.2. Interestingly, this automatic migration service yields the biggest increase in consumer surplus (24.8%), total welfare (21.1%) and cloud usage (21.9%) relative to the baseline. The provider revenue, net of the costs of the service, rises by 12.2%, which is lower than those under the personalized product trials but higher than those under the new product previews. Therefore, even though it is not profitable for the provider to pay for the automatic migration service during our sample period, it might be profitable in the long run.

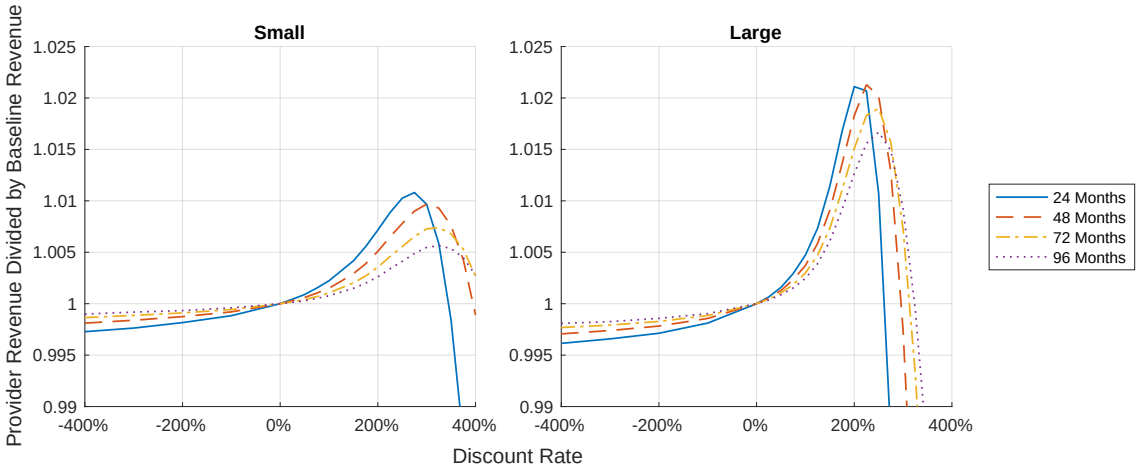
E.1 Optimal Introductory Discounts

In this section, we introduce how we find the provider's optimal introductory discounts and compare the short-run vs. long-run optimal discount rates. For each type of discount (new product preview or personalized product trial), we conduct a grid search over discount rates from -1000% to 1000%, where a discount rate of x on price p_{jm} implies a discounted price of $(1-x)p_{jm}$. We first conduct a coarse search over this range with 100% intervals to narrow down to a 200% range for the optimal discount. Then, within this 200% range, we refine the grid search to 25% intervals. For each discount, we fix characteristics of each market (customers, choice sets, and prices) and simulate demand forward following the same procedure as in Section 6.1 to compute consumer surplus, provider revenue, and cloud usage. We average across 1000 simulations to find the provider's optimal discount rates.

To highlight the difference between the short-run vs. long-run trade-off for the provider, in Figures E1 and E2, we present one line each representing the relationship between the

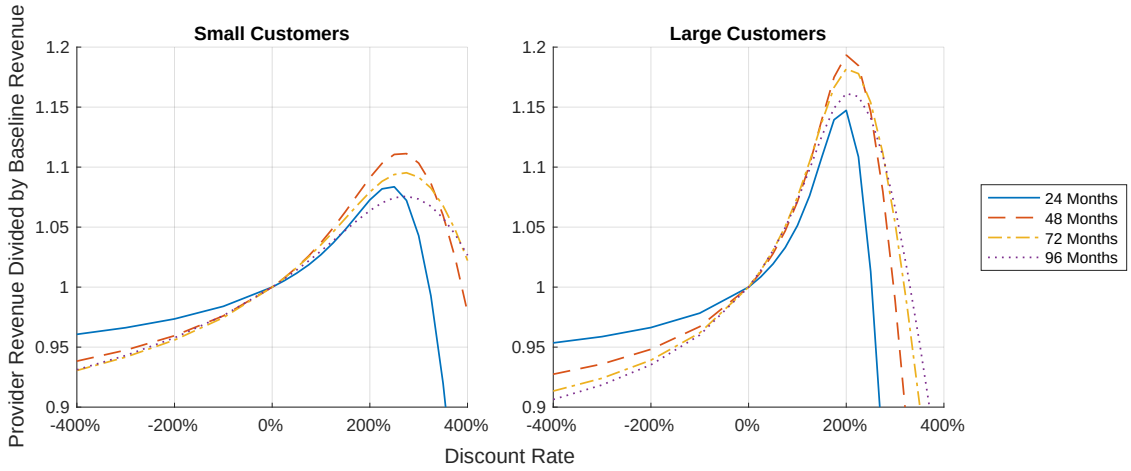
introductory discount rates and the provider's revenue, for when the provider's total revenue objective is defined for 24, 48, 72, and 96 months after X4's launch. The peak of each curve represents the corresponding optimal discount. As the provider's horizon becomes longer, the optimal discount becomes larger as the benefit of overcoming customer inertia increases. The optimal discount stabilizes between 72 and 96 months as later months do not contribute as much to the provider's objective.

Figure E1: New product preview: Different horizons



Notes: Figure shows the ratio of provider revenue with different introductory discounts relative to the baseline, separately for small and large customers. Different lines represent different horizons in the provider's revenue calculations. Results are averages across 1000 simulations. Discount rates, shown on the x-axis, are in absolute levels: a discount rate of x means that, in the first month after X4's launch, customers would pay (1-x) times their original spending on X4.

Figure E2: Personalized product trial: Different horizons



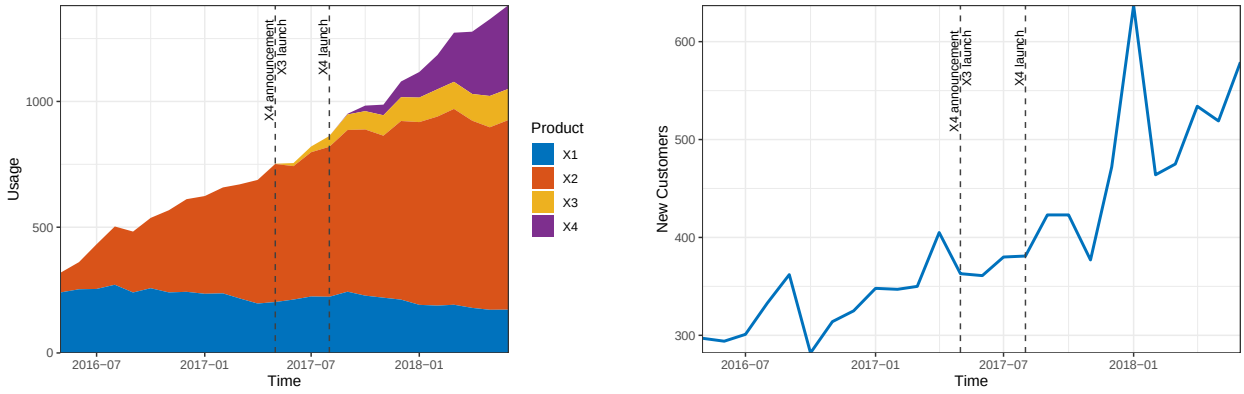
Notes: Figure shows the ratio of provider revenue with different introductory discounts relative to the baseline, separately for small and large customers. Different lines represent different horizons in the provider's revenue calculations. Results are averages across 1000 simulations. Discount rates, shown on the x-axis, are in absolute levels: a discount rate of x means that, in the first month that a customer uses X4, the customer would pay (1-x) times its original spending on X4.

F Additional Figures and Tables

Figure F3: Total usage within X series and number of new customers

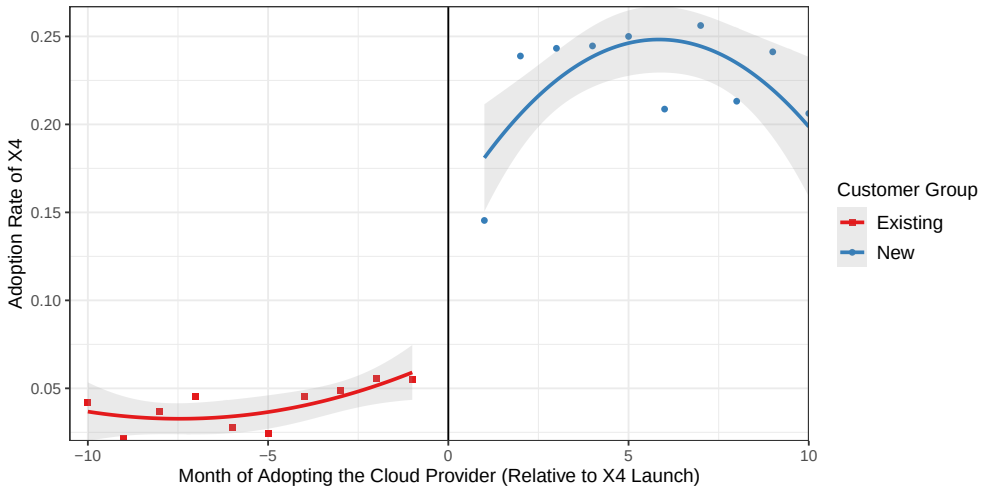
(a) Usage of X series products

(b) Number of new customers



Notes: Panel (a) plots total monthly usage of X1–X4 for all customers by product. Panel (b) plots the number of new customers joining the cloud provider each month. The first dashed vertical line marks the X3 launch and the announcement of forthcoming X4 launch; the second dashed vertical line marks the X4 launch. When X3 was launched, the forthcoming launch of X4 was publicly announced over three months in advance. Yet neither the cloud usage nor the customer base showed any obvious slowdown in growth during that period, as one might expect if customers were dynamically optimizing when to join the provider. One caveat is that the announcement coincided with the launch of X3, so this is not a definitive test. However, it provides some reassurance for the myopic assumption.

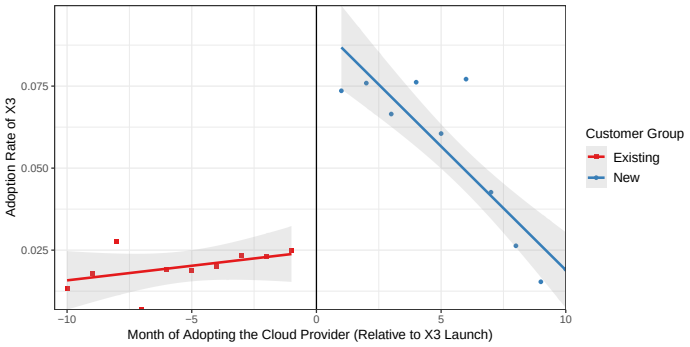
Figure F4: X4 adoption rate after its launch: regression discontinuity (quadratic control)



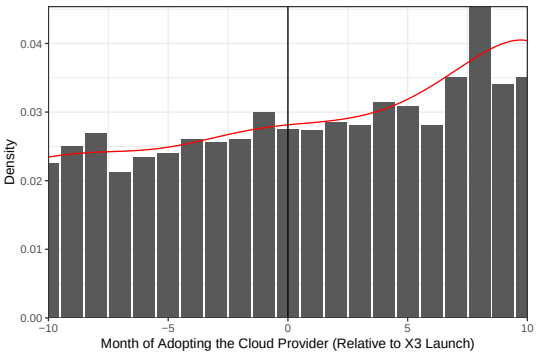
Notes: Figure plots the RDD comparing adoption rates of X4 between existing and new customers to the provider. The running variable is each customer's month of first adopting the provider relative X4's launch month. The outcome variable is a dummy indicating whether X4 is adopted during the month of X4's launch and the subsequent 10 months. We adopt the donut-RDD and restrict the sample to customers adopting the provider during the 10 months before and after X4's launch, excluding the month of the launch. Lines are regressions of the outcome variable on a quadratic function of the running variable before and after the discontinuity. Gray shaded area represents 95% confidence interval.

Figure F5: X3 adoption rate after its launch

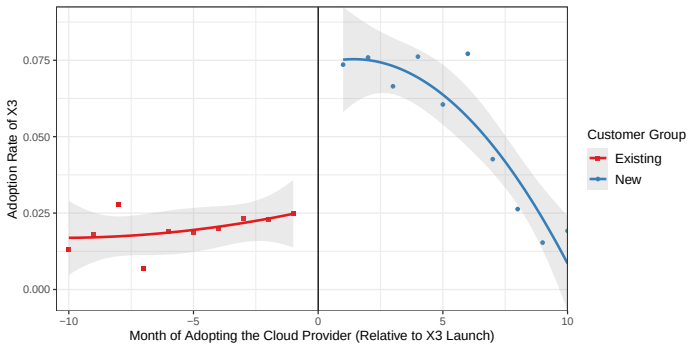
(a) Regression discontinuity (linear control)



(b) Density of the running variable

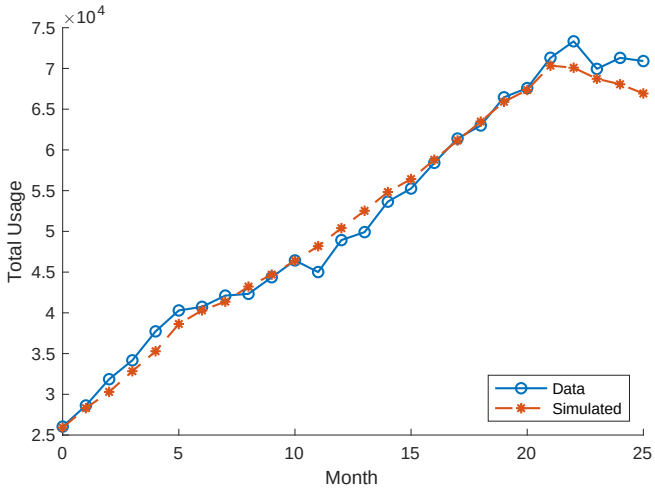


(c) Regression discontinuity (quadratic control)



Notes: Figure plots the RDD comparing adoption rates of X3 between existing and new customers to the provider. The running variable is each customer's month of first adopting the provider relative X3's launch month. The outcome variable is a dummy indicating whether X3 is adopted during the month of X3's launch and the subsequent 10 months. We adopt the donut-RDD and restrict the sample to customers adopting the provider during the 10 months before and after X3's launch, excluding the month of the launch. Panel (a) and (c) present the RDD results. Lines are regressions of the outcome variable on the running variable in panel (a) and on a quadratic function of the running variable in panel (c) before and after the discontinuity. Gray shaded area represents 95% confidence interval. Panel (b) presents the density of the running variable. The red line is the fitted kernel density of the running variable. The gray bars are the raw frequencies (rescaled as the probability densities) of the running variables. The post-launch adoption rate of X3 declines over time because some new customers shift onto X4 after X4 is launched.

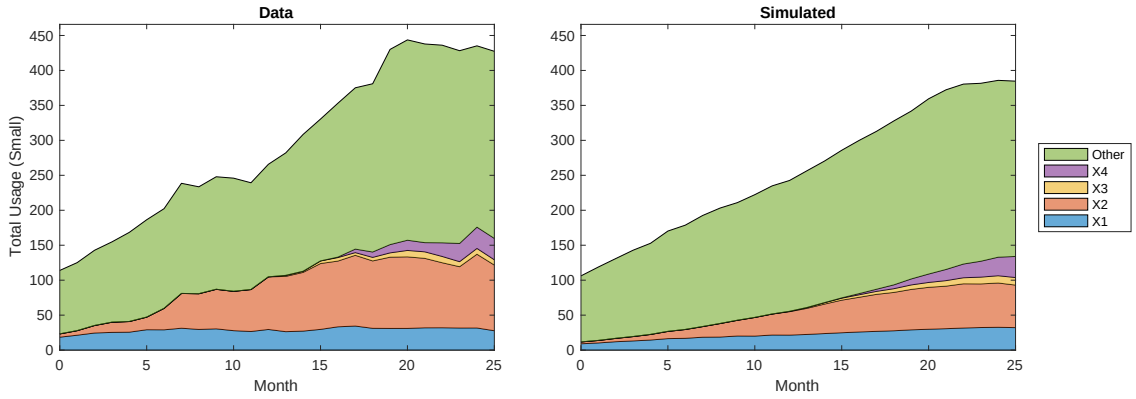
Figure F6: Time series fit of total computing demand



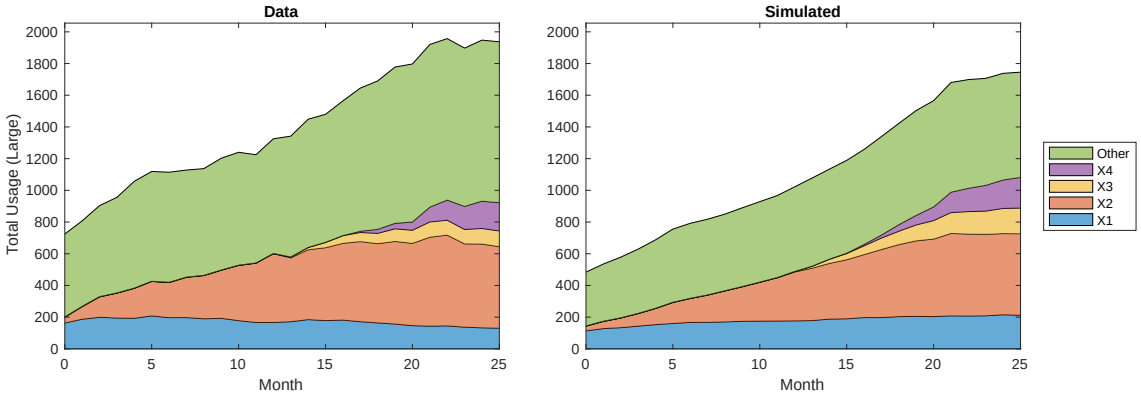
Notes: Figure shows total computing demand (including outside option) over time (months from beginning of sample) from the data (blue solid line with circles) and from simulations at our estimates (red dashed line with stars).

Figure F7: Time series fit of cloud demand by product and customer size

(a) Small Customers



(b) Large Customers



Notes: Figure shows cloud usage by product over time (months since beginning of sample) from the data (left) and from simulations at our estimates (right). Figure (a) shows the usage for small customers and figure (b) shows the usage for large customers.

Table F7: Summary statistics

Variable	N	Mean	SD	1st quartile	Median	3rd quartile	Max
Price, per product-market-month	3755	0.565	0.213	0.371	0.573	0.732	1.079
Market share, per product-market-month	3755	0.282	0.273	0.050	0.190	0.481	1
Number of products, per market-month	1209	3.106	1.446	3	3	4	5
Usage, per customer-market-month	96878	0.463	1.447	0	0.120	0.465	75.966

Notes: Table shows summary statistics of our estimation sample. Market is defined as a region-OS combination. Products include X1–X4, and the "other" product. Price is measured in re-scaled dollars per compute unit. Market share represents the usage share of each product, excluding the outside option, in each market-month. Usage is measured in re-scaled compute units.

Table F8: Customer cohorts' usage shares within X series (no attrition)

	(A) Before X3	(B) New Customer Between X3 to X4	(C) New Customer After X4
(a) Before X3 Usage Share			
X1	42%	-	-
X2	58%	-	-
(b) X3 to X4 Usage Share			
X1	27%	14%	-
X2	71%	68%	-
X3	1%	18%	-
(c) After X4 Usage Share			
X1	23%	11%	10%
X2	64%	66%	54%
X3	5%	19%	7%
X4	7%	4%	30%

Notes: Table presents usage share across products within X series for three customer cohorts and three time periods. The three cohorts (presented in columns) are (A) customers who adopted the cloud provider before the launch of X3, (B) customers who adopted the cloud provider between the launch of X3 and X4, and (C) customers who adopted the cloud provider after the launch of X4. The three time periods are (a) before the launch of X3, (b) between the launch of X3 and X4, and (c) after the launch of X4. Here we only keep customers who continue to use the X series in all subsequent time periods once they join the cloud provider, i.e., there is no attrition. Panel (a)–(c) present the usage shares in each of the three time periods.

Table F9: Regression discontinuity result at launch of X4

	No Control	Linear	Quadratic
Estimated Gap	0.18	0.15	0.08
	(0.01)	(0.02)	(0.03)
N	7547	7547	7547

Notes: Table presents the RDD estimates comparing adoption rates of X4 between existing and new customers to the provider. The running variable is each customer's month of first adopting the provider relative X4's launch month. The outcome variable is a dummy indicating whether X4 is adopted during the month of X4's launch and the subsequent 10 months. We adopt the donut-RDD and restrict the sample to 10 months before and after the launch of X4, excluding the month of launch itself. Column 1 includes no control; column 2 controls for a linear function of the running variable; column 3 controls for a quadratic function of the running variable. Standard errors in parentheses are clustered at the market-running variable level.

Table F10: Regression discontinuity result at launch of X3

	No Control	Linear	Quadratic
Estimated Gap	0.03	0.07	0.05
	(0.00)	(0.01)	(0.02)
N	7696	7696	7696

Notes: Table presents the RDD estimates comparing adoption rates of X3 between existing and new customers to the provider. The running variable is each customer's month of first adopting the provider relative X3's launch month. The outcome variable is a dummy indicating whether X3 is adopted during the month of X3's launch and the subsequent 10 months. We adopt the donut-RDD and restrict the sample to 10 months before and after the launch of X3, excluding the month of launch itself. Column 1 includes no control; column 2 controls for a linear function of the running variable; column 3 controls for a quadratic function of the running variable. Standard errors in parentheses are clustered at the market-running variable level.

Table F11: RDD with different widths of the window

Window (months)	Estimate - X3	Estimate - X4
5	0.052 (0.014)	0.096 (0.025)
7	0.054 (0.012)	0.129 (0.021)
10 (baseline)	0.070 (0.009)	0.154 (0.018)

Notes: Table presents the RDD estimates with different width of the window, restricting the sample to customers who joined the provider within 5, 7, and 10 months on either side of the product launch. Column 1 presents the estimates for the X3 launch and column 2 presents the estimates for the X4 launch, controlling for a linear function of the running variable. Standard errors in parentheses are clustered at the market-running variable level.

Table F12: RDD with different adoption horizons

Horizon (months)	Estimate - X3	Estimate - X4
1	0.046 (0.011)	0.086 (0.015)
2	0.057 (0.022)	0.029 (0.037)
3	0.069 (0.019)	0.066 (0.028)
4	0.058 (0.014)	0.090 (0.024)
5	0.058 (0.012)	0.118 (0.023)
6	0.056 (0.012)	0.142 (0.021)
7	0.060 (0.011)	0.138 (0.019)
8	0.067 (0.010)	0.145 (0.019)
9	0.069 (0.010)	0.149 (0.019)
10 (baseline)	0.070 (0.009)	0.154 (0.018)

Notes: Table presents the RDD estimates with different horizons over which we define $AdoptNew_{ij}$. Column 1 presents the estimates for the X3 launch and column 2 presents the estimates for the X4 launch, controlling for a linear function of the running variable. Standard errors in parentheses are clustered at the market-running variable level.

Table F13: RDD without transient customers

Product launch	Tenure threshold (active months)	Estimate
X3	0 (baseline)	0.070 (0.009)
X4	0 (baseline)	0.154 (0.018)
X3	3	0.095 (0.017)
X4	3	0.123 (0.033)
X3	6	0.089 (0.025)
X4	6	0.030 (0.047)

Notes: Table presents the RDD estimates where we exclude customers with fewer months of cloud usage than certain tenure thresholds. The baseline RDD uses the entire sample. The estimation sample excludes transient customers with fewer than 6 months of tenure. We control for a linear function of the running variable. Standard errors in parentheses are clustered at the market-running variable level. The RDD estimate is imprecise for the launch of X4 once we exclude customers with fewer than 6 months of tenure, because we exclude 90% of *New* customers.

Table F14: Robustness: Demand estimates identifying inertia with only RDD moments

Parameter	Estimate	SE	Estimate	SE
Panel (a) Product choice: means	Small		Large	
X1 (β_i^1)	-5.191	(0.462)	-2.995	(0.443)
X2/X3 (β_i^2)	-3.955	(0.267)	-3.071	(0.235)
X4 (β_i^4)	-4.001	(0.317)	-2.667	(0.279)
Other (β_i^O)	-2.492	(6.632)	-2.324	(6.660)
Promo (β_i^P)	-1.009	(0.249)	-0.124	(0.128)
Price $\times$ task size (α_i)	-8.684	(3.279)	-4.125	(2.262)
Adoption cost (δ_i)	-4.713	(0.979)	-10.709	(56.506)
Control function (κ_i)	2.571	(0.939)	1.196	(0.453)
Panel (b) Product choice: standard deviations				
X1 (σ^1)			0.084	(2.366)
X2 (σ^2)			0.911	(0.260)
X3 (σ^3)			0.894	(0.399)
X4 (σ^4)			0.441	(0.419)
Other (σ^O)			0.002	(38.256)
Panel (c) Number of tasks	Small		Large	
Intercept (λ_{i0})	45.329	(5.855)	75.027	(17.188)
Time trend (λ_{i1})	1.940	(0.273)	0.427	(0.159)
Panel (d) Task size	Small		Large	
Exponential scale (γ_i)	0.101	(0.013)	0.399	(0.091)

Notes: Table presents estimates of demand parameters. Standard errors are in parentheses and computed using estimated GMM asymptotic variance. We allow all parameters to depend on whether a customer is small or large, except for the standard deviations for the random coefficients in Panel (b). In Panel (a), (c), and (d), the first (last) two columns present the parameter estimates and standard errors for small (large) customers.

Table F15: Welfare robustness: inertia as dis-utility vs inertia choice-only

		Inertia as dis-utility Consumer Surplus (10^3)		**Inertia choice-only Consumer Surplus (10^3)**	
		(Small)	(Large)	(Small)	(Large)
Baseline: with inertia	By size	13.2	38.2	13.7	39.2
		(0.1)	(0.3)	(0.1)	(0.3)
	Total	51.5		53.0	
		(0.3)		(0.4)	
Counterfactual: without inertia	By size	33.0	103.3	33.0	103.3
		(0.2)	(0.5)	(0.2)	(0.5)
	Total	136.3		136.3	
		(0.5)		(0.5)	
Δ Welfare (without - with)	By size	19.8	65.1	19.3	64.0
		(0.1)	(0.4)	(0.1)	(0.5)
	Total	84.8		83.3	
		(0.5)		(0.5)	

Notes: Table reports a comparison of the welfare cost of inertia under two treatments of inertia in the consumer surplus calculation. In the first ("inertia as dis-utility"), inertia reduces consumer utility. In the second ("inertia choice-only"), inertia affects choices but is not counted in consumer utility, so it does not reduce consumer surplus. Table shows consumer surplus in the estimated model (baseline, with inertia) and in a counterfactual where inertia is eliminated, with the implied welfare cost given by the difference. Consumer surplus is shown in re-scaled dollars. Results are based on 200 simulations, with standard deviations in parentheses.